



FINAL YEAR PROJECT INTERIM REPORT

FYP01-SE-T2530-0794  
Event Management System with Automated Scheduling Optimization  
Using Genetic Algorithms

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Bachelor of Computer Science (Hons.) (Software Engineering)

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**Event Management System with Automated Scheduling**  
**Optimization Using Genetic Algorithms**

BY

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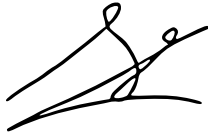
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## **Abstract**

At Multimedia University (MMU), event management is often done manually and with fragmented methods. Information regarding events is shared through multiple platforms, while each case of venue booking and allocation have to be checked one by one by an administrator. This makes the process time-consuming and highly prone to human error. As a solution to this, the author intends to develop an Event Management System (EMS) which employs Genetic Algorithms (GA) to assist in event scheduling optimization. This would include a centralized location for searching and registering for events, automated optimized event scheduling and venue allocation. To elicit requirements for the proposed system, a survey was conducted alongside several interviews with university members including students, committee members, and event organizers. Upon implementation, predicted benefits to current event management processes include reduced scheduling conflicts, more efficient venue usage, and an overall better user experience for university members of all roles.

## Table of Contents

|                                                 |      |
|-------------------------------------------------|------|
| Copyright.....                                  | v    |
| Declaration.....                                | vi   |
| Acknowledgements .....                          | vii  |
| Abstract.....                                   | viii |
| Table of Contents.....                          | ix   |
| List of Tables .....                            | xii  |
| List of Figures .....                           | xiii |
| List of Abbreviations/Symbols .....             | xv   |
| List of Appendices .....                        | xvi  |
| Chapter 1: Introduction.....                    | 1    |
| 1.1 Overview .....                              | 1    |
| 1.2 Problem Statement.....                      | 1    |
| 1.3 Objectives .....                            | 1    |
| 1.4 Project Scope.....                          | 2    |
| 1.5 Project Limitations .....                   | 2    |
| 1.6 Methodology .....                           | 3    |
| 1.7 Target Audience .....                       | 3    |
| 1.8 Summary.....                                | 3    |
| Chapter 2: Literature Review .....              | 5    |
| 2.1 Overview .....                              | 5    |
| 2.2 Schedule management.....                    | 5    |
| 2.2.1 Challenges with scheduling .....          | 6    |
| 2.3 Automated Scheduling .....                  | 6    |
| 2.4 Genetic Algorithms (GA).....                | 7    |
| 2.5 Existing Systems for Event Management ..... | 7    |
| 2.6 Summary.....                                | 8    |
| Chapter 3: Requirements Analysis.....           | 10   |
| 3.1 Overview .....                              | 10   |

|                                                              |    |
|--------------------------------------------------------------|----|
| 3.2 Fact-Finding Techniques .....                            | 10 |
| 3.2.1 Justification .....                                    | 10 |
| 3.2.2 Questionnaire Design .....                             | 10 |
| 3.2.3 Interview .....                                        | 25 |
| 3.3 Requirements.....                                        | 32 |
| 3.3.1 Functional Requirements.....                           | 32 |
| 3.3.1.1 Login Function .....                                 | 33 |
| 3.3.1.2 Event Discovery Function.....                        | 33 |
| 3.3.1.3 Registration Function.....                           | 33 |
| 3.3.1.4 Notification Function .....                          | 33 |
| 3.3.1.5 Scheduling Function .....                            | 33 |
| 3.3.1.6 Attendance Function.....                             | 33 |
| 3.3.1.7 Venue Management Function .....                      | 34 |
| 3.3.1.8 Event Planning Function.....                         | 34 |
| 3.3.1.9 Event Analytics Function .....                       | 34 |
| 3.3.2 Non-Functional Requirements .....                      | 34 |
| 3.3.2.1 Usability.....                                       | 34 |
| 3.3.2.2 Performance .....                                    | 34 |
| 3.3.2.3 Reliability.....                                     | 34 |
| 3.3.2.4 Availability .....                                   | 35 |
| 3.3.2.5 Security .....                                       | 35 |
| 3.3.2.6 Maintainability .....                                | 35 |
| 3.3.2.7 Scalability.....                                     | 35 |
| 3.3.2.8 Compatibility .....                                  | 35 |
| 3.3.3 User Requirements .....                                | 35 |
| 3.3.3.1 Centralized Event List.....                          | 35 |
| 3.3.3.2 Standardized and Simplified Event Registration ..... | 35 |
| 3.3.3.3 Event Reminder and Update Notification.....          | 35 |
| 3.3.3.4 Event Tracking and History .....                     | 35 |
| 3.3.3.5 Standardized Event Planning Workflow.....            | 36 |
| 3.3.3.6 Venue Availability Checker .....                     | 36 |
| 3.3.3.7 Participant Attendance Tracking.....                 | 36 |



|                                |                                          |      |
|--------------------------------|------------------------------------------|------|
| 3.3.3.8                        | Visibility Venue Usage and Conflict..... | 36   |
| 3.3.3.9                        | Standardize Event Documentation.....     | 36   |
| 3.3.3.10                       | Attendance Analytics .....               | 36   |
| 3.4                            | Summary.....                             | 36   |
| Chapter 4: System Design ..... |                                          | 36   |
| 4.1                            | Overview .....                           | 37   |
| 4.2                            | Rich Picture Diagram.....                | 37   |
| 4.3                            | Use Case Diagram .....                   | 37   |
| 4.4                            | Activity Diagram.....                    | 39   |
| 4.5                            | Class Diagram .....                      | 47   |
| 4.6                            | Sequence Diagram .....                   | 48   |
| 4.7                            | Interface Design.....                    | 53   |
| 4.8                            | Summary.....                             | 60   |
| Chapter 5: Implementation..... |                                          | 60   |
| 5.1                            | Activities in FYP2.....                  | 60   |
| 5.2                            | FYP2 Milestones.....                     | 60   |
| 5.2.1                          | Milestones.....                          | 60   |
| Chapter 6: Conclusion .....    |                                          | 61   |
| References.....                |                                          | xii  |
| Appendix A.....                |                                          | xiii |
| Appendix B .....               |                                          | xliv |
| Appendix C .....               |                                          | xlvi |

List of Tables

Table 5.1: Milestones for FYP2 ..... 60

## List of Figures

|                                                                                |    |
|--------------------------------------------------------------------------------|----|
| Figure 3.1: Respondents' Age Group.....                                        | 16 |
| Figure 3.2: Respondents' University Roles.....                                 | 17 |
| Figure 3.3: Likelihood of Respondents to Use EMS .....                         | 17 |
| Figure 3.4: Most Common Channels for students to Learn about Events.....       | 18 |
| Figure 3.5: Difficulty of Event Registration.....                              | 19 |
| Figure 3.6: Problems with Event Management Processes .....                     | 19 |
| Figure 3.7: Respondents' Sources of Influence for Joining Events .....         | 20 |
| Figure 3.8: Problems Respondents Encounter with University Events .....        | 21 |
| Figure 3.9: Respondents' Desired Features for an EMS .....                     | 21 |
| Figure 3.10: Committee members' Preferred Event Management Tools.....          | 22 |
| Figure 3.11: Effectiveness of committee members' Workflow.....                 | 22 |
| Figure 3.12: Challenges of Committee Event Management .....                    | 23 |
| Figure 3.13: Tools that would Make Event Planning Easier .....                 | 24 |
| Figure 3.14: Period of Event Planning for committee members .....              | 24 |
| Figure 3.15: Smoothness of Venue Booking Process .....                         | 25 |
| Figure 3.16: Frequency of Difficulties When Acquiring Approval for Event ..... | 25 |
| Figure 4.1: Current Event Management Process in MMU .....                      | 37 |
| Figure 4.2: Use Case Diagram of EMS .....                                      | 38 |
| Figure 4.3: Login for Users.....                                               | 39 |
| Figure 4.4: Event Registration for Students.....                               | 40 |
| Figure 4.5: Event Organizers Updating Event Details.....                       | 41 |
| Figure 4.6: Event Organizers Making Announcements.....                         | 42 |
| Figure 4.7: Event Organizers Creating Event Proposal .....                     | 43 |
| Figure 4.8: Event Organizers Generating Attendance QR Code .....               | 44 |
| Figure 4.9: Administrators Viewing Event Proposals.....                        | 45 |
| Figure 4.10: Administrators Viewing Venue Usage .....                          | 46 |
| Figure 4.11: Administrators Viewing Event Analytics/Report.....                | 47 |
| Figure 4.12: EMS Class Diagram .....                                           | 48 |
| Figure 4.13: User Login Process .....                                          | 49 |
| Figure 4.14: Student Event Search Process .....                                | 50 |
| Figure 4.15: Student Event Registration Process .....                          | 50 |
| Figure 4.16: Student Notification or Reminder Flow.....                        | 51 |
| Figure 4.17: Student In-app Calendar .....                                     | 51 |
| Figure 4.18: QR Code Event Attendance .....                                    | 52 |
| Figure 4.19: Venue Request and Admin Approval.....                             | 52 |
| Figure 4.20: Event Analytics Generation .....                                  | 53 |
| Figure 4.21: EMS Login Page.....                                               | 53 |
| Figure 4.22: EMS Register Page .....                                           | 54 |

|                                                           |    |
|-----------------------------------------------------------|----|
| Figure 4.23: EMS Student Dashboard .....                  | 54 |
| Figure 4.24: EMS Student Event Search.....                | 55 |
| Figure 4.25: EMS Student Personal Events List .....       | 55 |
| Figure 4.26: EMS Student Personal Schedule .....          | 56 |
| Figure 4.27: EMS Student Notifications List.....          | 56 |
| Figure 4.28: EMS Organizer Dashboard .....                | 57 |
| Figure 4.29: EMS Organizer Event Planning Dashboard ..... | 57 |
| Figure 4.30: EMS Organizer Event Creation Window .....    | 58 |
| Figure 4.31: EMS Admin Dashboard .....                    | 58 |
| Figure 4.32: EMS Admin Analytics Dashboard.....           | 59 |

## **List of Abbreviations/Symbols**

AI Artificial Intelligence

EMS Event Management System

FYP2 Final Year Project 2

GA Genetic Algorithm

IRJET International Research Journal of Engineering and Technology

MMU Multimedia University

QR Quick Response

UI User Interface

## List of Appendices

|                                                                |          |
|----------------------------------------------------------------|----------|
| Appendix A 1: Meeting Log 1 Page 1.....                        | xiii     |
| Appendix A 2: Meeting Log 1 Page 2.....                        | xiv      |
| Appendix A 3: Meeting Log 1 Page 3.....                        | xv       |
| Appendix A 4: Meeting Log 1 Page 4.....                        | xvi      |
| Appendix A 5: Meeting Log 1 Page 5.....                        | xvii     |
| Appendix A 6: Meeting Log 1 Page 6.....                        | xviii    |
| Appendix A 7: Meeting Log 2 Page 1.....                        | xix      |
| Appendix A 8: Meeting Log 2 Page 2.....                        | xx       |
| Appendix A 9: Meeting Log 2 Page 3.....                        | xxi      |
| Appendix A 10: Meeting Log 2 Page 4.....                       | xxii     |
| Appendix A 11: Meeting Log 2 Page 5.....                       | xxiii    |
| Appendix A 12: Meeting Log 2 Page 6.....                       | xxiv     |
| Appendix A 13: Meeting Log 3 Page 1.....                       | xxv      |
| Appendix A 14: Meeting Log 3 Page 2.....                       | xxvi     |
| Appendix A 15: Meeting Log 3 Page 3.....                       | xxvii    |
| Appendix A 16: Meeting Log 3 Page 4.....                       | xxviii   |
| Appendix A 17: Meeting Log 3 Page 5.....                       | xxix     |
| Appendix A 18: Meeting Log 4 Page 1.....                       | xxx      |
| Appendix A 19: Meeting Log 4 Page 2.....                       | xxxi     |
| Appendix A 20: Meeting Log 4 Page 3.....                       | xxxii    |
| Appendix A 21: Meeting Log 4 Page 4.....                       | xxxiii   |
| Appendix A 22: Meeting Log 4 Page 5.....                       | xxxiv    |
| Appendix A 23: Meeting Log 5 Page 1.....                       | xxxv     |
| Appendix A 24: Meeting Log 5 Page 2.....                       | xxxvi    |
| Appendix A 25: Meeting Log 5 Page 3.....                       | xxxvii   |
| Appendix A 26: Meeting Log 5 Page 4.....                       | xxxviii  |
| Appendix A 27: Meeting Log 5 Page 5.....                       | xxxix    |
| Appendix A 28: Meeting Log 6 Page 1.....                       | xl       |
| Appendix A 29: Meeting Log 6 Page 2.....                       | xli      |
| Appendix A 30: Meeting Log 6 Page 3.....                       | xlii     |
| Appendix A 31: Meeting Log 6 Page 4.....                       | xliii    |
| Appendix A 32: Meeting Log 6 Page 5.....                       | xliv     |
| <br>Appendix B 1: Turnitin Similarity Index .....              | <br>xlv  |
| <br>Appendix C 1: FYP Student Interview Transcript Page 1..... | <br>xlvi |
| Appendix C 2: FYP Student Interview Transcript Page 2.....     | xlvii    |
| Appendix C 3: FYP Student Interview Transcript Page 3.....     | xlviii   |
| Appendix C 4: FYP Student Interview Transcript Page 4.....     | xlix     |

|                                                                     |      |
|---------------------------------------------------------------------|------|
| Appendix C 5: FYP Student Interview Transcript Page 5.....          | I    |
| Appendix C 6: FYP Student Interview Transcript Page 6.....          | li   |
| Appendix C 7: FYP Student Interview Transcript Page 7.....          | lii  |
| Appendix C 8: FYP Event Organizer Interview Transcript Page 1.....  | liii |
| Appendix C 9: FYP Event Organizer Interview Transcript Page 2.....  | liv  |
| Appendix C 10: FYP Event Organizer Interview Transcript Page 3..... | lv   |

## **Chapter 1: Introduction**

### **1.1 Overview**

At Multimedia University (MMU), event management is often done manually whereby activities such as event scheduling and venue allocation are not automated and instead done by hand. This is obviously very inefficient and often leads to the occurrence of scheduling conflicts. For this reason, the author proposes the implementation of an Event Management System (EMS) that employs Genetic Algorithms (GA) to automate event scheduling optimization, taking into account factors such as venue capacity, availability and organizer preferences. Employing the Kanban methodology, the system will be developed incrementally with the tasks performed all displayed on a Kanban board. Upon completion, the expected outcome of this project is a web-based prototype that generates conflict-free and optimized event schedules which will greatly reduce the manual effort required to perform event management tasks.

### **1.2 Problem Statement**

Current event scheduling processes at universities heavily rely on coordination between various manual tasks to ensure venue availability, date alignment, and non-conflicting scheduling. This results in high chances of scheduling conflicts, especially at near-maximum student intake, inefficient utilization of venues, where organizers may fail to take note of more appropriate venue allocation options, and time-consuming planning, as organizers would have to make many repeated checks with relevant parties such as venue managers and administrator staff.

Without automation or optimization, universities face increased administrative load and reduced operational efficiency. Therefore, a system that automates scheduling and optimizes timetable generation is needed to address these limitations.

### **1.3 Objectives**



The objectives of this project are as follows:

1. To design and develop an Event Management System (EMS) integrated with a scheduling optimization module based on Genetic Algorithms (GA).
2. To implement a fitness evaluation function that optimizes event–venue– time allocations according to defined constraints such as capacity, availability, and preference.
3. To test and evaluate the system's performance in minimizing scheduling conflicts, maximizing venue utilization, and assessing user satisfaction and usability among event organizers.

### **1.4 Project Scope**

The scope of this project includes an EMS that can handle event registration, venue and date management while making use of GA-based schedule optimization for venue, time slot, and capacity. It will also include a basic admin/organizer dashboard for schedule visualization, and a comparison report of automated vs manual scheduling.

Upon completion, this project will result in a functional prototype for a web-based EMS that is capable of optimizing event schedules with the use of a GA module. To show and prove the need for such a system, an evaluation report comparing automated and manual scheduling efficiency will be made.

### **1.5 Project Limitations**

This project has the following limitations: the system will function as a prototype, not a full enterprise solution; only venue capacity, availability, blackout dates, and organizer preferences are considered. More advanced constraints such as event type dependencies and multi-day events are not included; GA optimization is designed for moderate-scale datasets and may not be fully optimized for extremely large or complex scheduling environments; external system integrations such as calendar sync and payment gateways are not

included; the system will not be portable as it will be designed mainly for web use and use on mobile is unaddressed.

## **1.6 Methodology**

This project will be developed with the Kanban framework, a workflow management approach focused on continuous incremental feature delivery and visual task tracking. In order to elicit system requirements for the EMS before commencing development, the author will conduct a survey with a questionnaire made specifically for students, committee members, event organizers and administrators. The target number of responses is at least 30, with no priority given to the composition of the demographics of the respondents. For the purpose of substantiating the results of the survey, the author will also carry out interview sessions with a targeted number of five interviewees consisting of people from the same university roles mentioned above.

## **1.7 Target Audience**

The primary users and beneficiaries of the system include: administrators who would be responsible for configuring booking policies, managing master data, and monitoring schedule performance; Venue Managers who would handle venue profiles, availability, maintenance schedules, and conflict resolutions; Event organizers which include students, lecturers, staff, and committees who would submit event requests, review suggestions, and manage bookings; and Attendees who consist of students, staff, and members of the public who would be able to view published event schedules and participate in relevant activities. In this way, the system would be designed to support both operational users (administrators, Venue Managers) and end users (students, organizers).

## **1.8 Summary**

This chapter has introduced the motivation and background of the project, outlining the existing problems with manual event scheduling in universities. The project aims to address these problems using a GA-driven EMS developed with the Kanban methodology. The problem statement, objectives, scope, project

limitations, methodology, and target audience have been clearly defined to establish the foundation for subsequent chapters. In the next chapter, I will go over the sources which I will refer to ensure the project the development of the EMS proceeds with as little problems as possible.

## **Chapter 2: Literature Review**

### **2.1 Overview**

Scheduling plays a critical role in environments where multiple factors such as resources, time slots, and constraints must be taken into account and coordinated efficiently. Universities rely on effective scheduling to manage events, venues, and users. However, this is often done with traditional scheduling methods which are manual, time-consuming, and prone to human error. The literature highlights several approaches to scheduling, ranging from simple rule-based systems to advanced optimization methods such as GA. Existing EMS primarily focus on workflow and registration rather than automated conflict-free scheduling. This chapter reviews schedule management concepts, scheduling challenges, automated scheduling techniques, GA fundamentals, and current EMS to highlight gaps addressed by this project.

### **2.2 Schedule management**

Schedule management refers to the process of allocating resources over time to perform required tasks. Wolf (1991) emphasizes that scheduling systems must manage operations, processors (resources), decisions, and constraints in a structured, flexible manner, especially when schedules must change dynamically. Effective schedule management involves ensuring feasibility, preventing conflicts, and maintaining clarity over resource allocation. Object-oriented scheduling models allow systems to represent resources, constraints, and tasks hierarchically where more generic features of schedule management are clearly separated from more domain-specific ones. This improves the system's reusability, maintainability, and adaptability.

In regards to construction management, Parsamehr et al (2022) states that effective schedule management involves taking into account details such as resource and equipment allocation, time–cost trade-offs, constructability issues, and optimum productivity. Since scheduling determines the sequence of activities

to be performed, without proper schedule management, resources would not be efficiently allocated especially when working at limited site spaces.

### **2.2.1 Challenges with scheduling**

Scheduling is highly dependent on various factors such as time and resources (Wolf, 1991; Parsamehr et al, 2022), contributing to its complexity. This can be seen in the many challenges it poses in several aspects. For example, when resources such as venues are limited, scheduling is made more difficult due to the need to avoid schedule and venue overlapping. Dynamic changes which often occur in the real world also add to its difficulty as one small change can cause a chain of required changes. This is especially true with the eventuality of human error which is inevitable with manual scheduling with its time-consuming nature. All these point to the need for an automated system capable of schedule-optimizing even with the various constraints.

## **2.3 Automated Scheduling**

Automated scheduling methods aim to mitigate the inefficiencies of manual scheduling by introducing algorithms capable of resolving conflicts and optimizing allocations. A comprehensive review of scheduling automation research from 1985 to 2014 (Faghihi et al, 2015) shows that among numerous AI-based methods such as expert systems, case-based reasoning and model-based scheduling, GA emerged as the most dominant and successful approach for complex multi-constraint problems. Automated scheduling systems help by: generating conflict-free schedules; handling large search spaces efficiently; considering multiple objectives simultaneously (e.g., time, cost, quality); reducing manual intervention and administrative workload.

In an effort to create a more balanced rotation schedule which takes into account individual preferences and guidelines provided by the Accreditation Council for Graduate Medical Education, Howard et al (2020) designed a rotation scheduling tool which made use of automation. Comparing schedule uniformity,

fulfillment of resident preferences and conflicting shift assignments of schedules generated with the newly developed scheduling tool with those of the previous schedule that was hand-made, they found that there was a significant decrease in schedule conflicts from 0.7 per intern to 0.3. Rates of average satisfaction and perception of fairness also rose, being 0.7 and 0.9 points higher compared to the manual scheduling system. This proves the significance and advantages of automated scheduling compared to its non-manual counterpart.

Despite these advancements, many operational systems still rely on rule-based conflict checking rather than optimization algorithms. This gap creates opportunities for GA-based solutions.

## **2.4 Genetic Algorithms (GA)**

GA are optimization techniques inspired by natural selection (Kramer, 2017; Wei et al, 2024). A GA operates on a population of candidate solutions, evolving them across generations using selection, crossover, and mutation processes. Fitness functions evaluate how good each solution is in satisfying the problem's constraints and objectives. According to the GA Essentials document, the key components include: Chromosomes, encoded representations of solutions; Fitness Functions which measure how optimal or feasible a solution is; Selection Mechanisms which choose high-quality solutions for reproduction; Crossover & Mutation which introduce variation and improve solution quality over iterations.

GA is very commonly used due to its several feats including that: It handles complex constraints and large solution spaces well; It supports multi-objective optimization; It is flexible and adaptable to different scheduling models. These properties make GA one of the most widely used optimization techniques for scheduling problems.

## **2.5 Existing Systems for Event Management**

Existing EMS primarily focus on facilitating event registration, user authentication, information dissemination, and communication between stakeholders. The IRJET paper describes typical EMS features such as: User login and registration; Online event participation; administrator updates on events; Publishing results of event rounds; and Issuing participation certificates via email (Arsheen et al, 2019). However, the paper also highlights major limitations: Existing EMS solutions are largely manual in scheduling; They do not automate venue allocation or conflict detection; They focus on workflow and information sharing rather than optimization; Most current systems support event registration but not optimized event, venue, and time allocation, revealing a significant gap for AI-based scheduling solutions.

Examples of existing EMS include Snoogo and EventMobi (Shah et al, 2023). Swoogo is designed to accommodate events whether they are physical, hybrid or online. This is exemplified by how the system allows for there to be multiple personalized registration routes for different purposes. Swoogo also allows event organizers to determine access level to registration data through two-factor authentication. EventMobi employs scalable event management whereby all events no matter how many or numerous will be automatically sorted so that event organizers are able to focus more on the designing audience engagement materials rather than overly focus on managing event schedules and timetables. This coincides with these two EMS being industry leaders among activity management applications, showing the advantages of their features.

This supports the justification for an EMS enhanced with GA to automatically generate conflict-free, optimized timetables.

## **2.6 Summary**

Research on automated scheduling shows the many limits of manual scheduling. In particular, compared to its automated counterpart, it is slow, prone to error, and incapable of handling complex constraints in relation to things like time and resources. The research also shows the superiority of GA among

various AI approaches, with GA its ability to optimize large search spaces and generate conflict-free allocations. This makes GA highly suitable for solving event scheduling problems. This being the case, it is noteworthy that no existing EMS makes use of it. This underscores the need for a GA solution to event management in MMU.



## **Chapter 3: Requirements Analysis**

### **3.1 Overview**

To ensure the design and development of the EMS provides for the needs of its users, I have conducted a survey inquiring people of my target demographics including students, committee members or more broadly, event organizers to elicit system requirements for the EMS. As of writing, I have managed to obtain 30 responses to the survey.

In addition, I have also conducted an interview with five figures who are have had experience in event participation and event management. Each interview lasted and around 15 minutes and in each, I asked a set of open-ended questions.

### **3.2 Fact-Finding Techniques**

#### **3.2.1 Justification**

Requirement elicitation is a critical phase of the development of a system as it helps ensure that the system meets the actual needs of the users rather than presumed ones. In the context of the EMS, as this system is intended to be used by several different users, a number of fact-finding techniques had to be employed to obtain the true requirements of the system. For this project, the author made use of two, questionnaires and interviews.

Questionnaires were chosen as they allowed for efficient and relatively fast collection of data from a large number of respondents. This allowed the author to gather quantitative data regarding user experiences with current university event management processes.

Interviews were chosen as they allowed the author to take a more in depth view at problems with current event management processes that could not be captured with quantitative survey data. This made it possible for the author to find further information on what issues or challenges users had and in turn, what to avoid.

#### **3.2.2 Questionnaire Design**

#### **Section A: Demographic profile**

This section contains questions whose objective is to determine the respondent's university role, tech-savviness, and so on.

**Question 1:** How old are you?

**Question objective:** To determine respondent's age because different age groups would have different requirements.

**Question Type:** Multiple choice

**Available options:** "Under 18 years old", "18-24", "25-34", "35-44", "45-54", "Above 54 years old"

**Question 2:** What is your role within the university?

**Question objective:** To determine respondent's role within the university because different university roles would have different requirements.

**Question Type:** Multiple choice

**Available options:** "student", "Committee member", "organizer", "Lecturer", "Admin"

**Question 3:** How likely are you to use an Event Management System app if available?

**Question objective:** To determine likelihood of respondent to use an EMS.

**Question type:** Likert scale

**Available options:** "Very unlikely" (1) – "Very likely" (5)

## **Section B: For Students**

**Question 1:** How do you usually find out about events on campus?

**Question objective:** To inquire methods in which students find out about events. (Select all that apply)

**Question type:** Multiple choice (Multiple answers)

**Available options:** Instagram, WhatsApp, Posters, Friends, Official MMU portal, Others (please specify)

**Question 2:** How easy do you find the current process of registering for events?

**Question objective:** To assess difficulty of using current registration methods for events.

**Question type:** Likert scale

**Available options:** Very difficult (1) - Very easy (5)

**Question 3:** What problems do you usually face when registering for events?  
(Select all that apply)

**Question objective:** To inquire problems faced by students when registering for events.

**Question type:** Multiple choice (Multiple answers)

**Available options:** Registration link not centralized, Confusing instructions, Slow response/confirmation, Hard to track events I joined, No reminders, Others (please specify)

**Question 4:** What influences your decision to participate in an event?

**Question objective:** To inquire what motivates students to participate in an event. (Select all that apply)

**Question type:** Multiple choice (Multiple answers)

**Available options:** Time, Location, Free gifts/merch, Certificate, Friends joining, Relevance to studies, Organized by specific club, i-Cadet requirements

**Question 5:** What frustrates you most about current campus events?

**Question objective:** To inquire what frustrates students the most about campus events. (Select all that apply)

**Question type:** Multiple choice (Multiple answers)

**Available options:** Scheduling clashes, Poor communication, Lack of updates, Poor organization, Hard to track my participation, Others (please specify)

**Question 6:** Which features would you like to have in an Event Management app? (Select all that apply)

**Question objective:** To inquire what features students would like to have in an EMS application.

**Question type:** Multiple choice (Multiple answers)

**Available options:** View all university events in one place, Smart recommendations, Registration + E-ticket Event reminders, Push notifications, Event history & certificates, Feedback system, Event calendar syncing to personal timetable, Others (please specify)

**Section C: For Committee Members**

**Question 1:** How does your committee currently manage event planning?

**Question objective:** To inquire how committee members manage event planning activities.

**Question type:** Multiple choice

**Available options:** Google Forms / WhatsApp / Manual spreadsheet / University portal / Other

**Question 2:** How effective is the current workflow for planning?

**Question objective:** To inquire how effective current event management methods are for committee members.

**Question type:** Likert scale

**Available options:** Very ineffective (1) - Very effective (5)

**Question 3:** What are the main challenges your committee faces when planning an event?

**Question objective:** To inquire what challenges committee members face during event planning.

**Question type:** Multiple choice (multiple answers)

**Available options:** Venue booking issues, Hard to coordinate with members, Lack of centralized documentation, Difficulty tracking attendance, Last-minute schedule clashes, Budget tracking, Approvals too slow, Communication problems, Other

**Question 4:** Which tools would make event planning easier for your committee?

**Question objective:** To inquire what tools would make it easier for committee members to plan events.

**Question type:** Multiple choice (multiple answers)

**Available options:** Centralized event dashboard, Automated approval workflow, Real-time communication, Checklist/task management, Budget + Expense tracking, Venue availability checker, Automated participant attendance, Templates for proposals & reports, Other

**Question 5:** On average, how long does it take from planning an event to final approval? (in days/weeks)

**Question objective:** To inquire how long it takes from event planning until final approval of the event.

**Question type:** Open-ended

**Question 6:** Rate your experience with venue booking.

**Question objective:** To inquire how smooth committee members' experience with venue booking is.

**Question type:** Semantic scale

**Available options:** Very problematic (1) - Very smooth (5)

**Question 7:** Do you face difficulties getting approval from the university for events?

**Question objective:** To inquire whether committee members face difficulties when getting approval from the university for events.

**Question type:** Multiple choice

**Available options:** Yes (please explain in the next question) / No

**Question 8:** Explain the difficulties you face in getting approval for events.

**Question objective:** To inquire what difficulties committee members face when getting approval from the university for events.

**Question type:** Open-ended

## **Section D: For Organizers**

**Question 1:** How does your department currently process event approvals? (Select all that apply)

**Question objective:** To inquire how departments process event approvals.

**Question type:** Multiple choice (multiple answers)

**Available options:** Manual forms, Emails, Google Forms, University portal, Other

**Question 2:** How many events do you process on average per month?

**Question objective:** To inquire how many events departments process on average every month.

**Question type:** Open-ended

**Question 3:** What are the biggest challenges you face in managing university events? (Select all that apply)

**Question objective:** To inquire what challenges organizers face in managing university events.

**Question type:** Multiple choice (multiple answers)

**Available options:** Too many requests, Duplicate/clashing events, Manual paperwork, No real-time visibility of venue usage, Tracking attendance & reporting, Ensuring compliance with rules, Poor communication with clubs, Other

**Question 4:** Do you face issues with venue double-booking?

**Question objective:** To inquire if organizers face issues with venue double-booking.

**Question type:** Multiple choice

**Available options:** Yes, No

**Question 5:** Which analytics would help you the most? (Select all that apply)

**Question objective:** To inquire what organizers would find the most helpful.

**Question type:** Multiple choice (multiple answers)

**Available options:** Event popularity, student engagement metrics, Venue utilization, Budget summaries, Club performance, Event success rating, Other

**Question 6:** What features are most essential in an Event Management System for admin-level users? (Select all that apply)

**Question objective:** To inquire what features organizers find to be the most essential.

**Question type:** Multiple choice (multiple answers)

**Available options:** Automated approvals, Role-based access, Document/proposal review, Participant verification, Attendance reporting, Risk management templates, Real-time venue availability, Audit logs, Conflict detection (schedule/venue), Other

**Question 7:** Would your department be open to adopting an Event Management System if it reduces manual workloads?

**Question objective:** To likelihood of adoption of EMS among departments.

**Question type:** Multiple choice

**Available options:** Yes / No / Maybe

**Question 8:** What concerns do you have about implementing an Event Management System?

**Question objective:** To inquire concerns organizers have with implementing a new EMS.

**Question type:** Open-ended

### 3.2.2.1 Analysis on Results

#### Section A: Demographics of Respondents

Over the course of two weeks, the author received 30 responses for the survey. In terms of age, 90% of the respondents were aged between 18-24 while the other 10% were of the ages between 25-34 as seen in Figure 3.1.

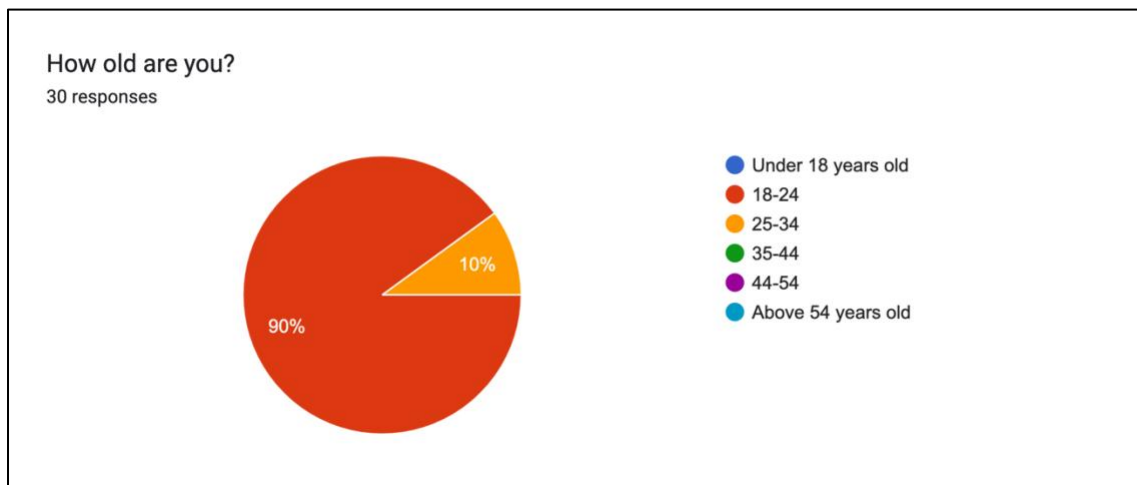


Figure 3.1: Respondents' Age Group

In terms of their roles within the university, 10% of the respondents were committee members while all the other respondents were students as seen in Figure 3.2.

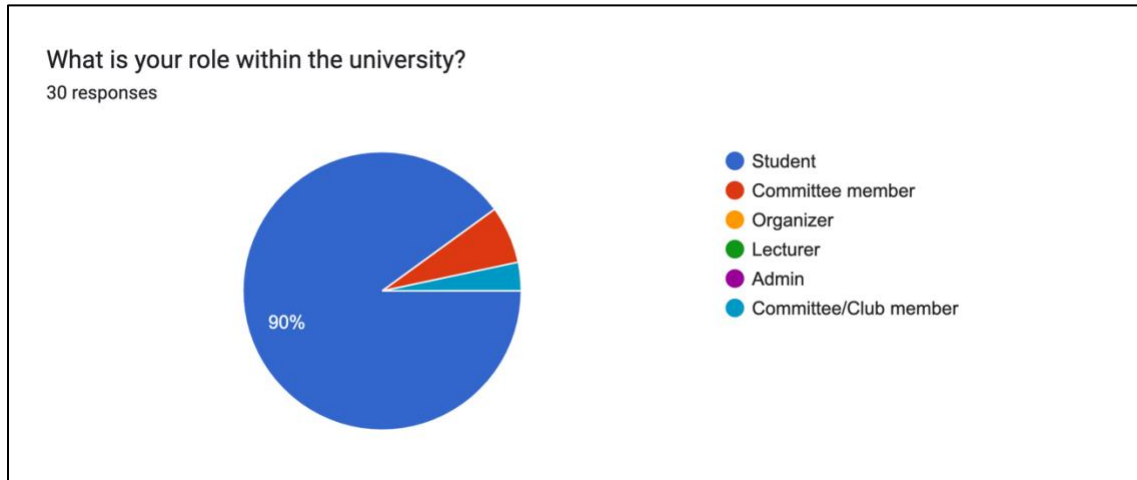


Figure 3.2: Respondents' University Roles

Overall, more than half of the respondents (63%) answered they were likely or very likely to use an EMS if available while only 10% responded they were unlikely to do so as seen in Figure 3.3.

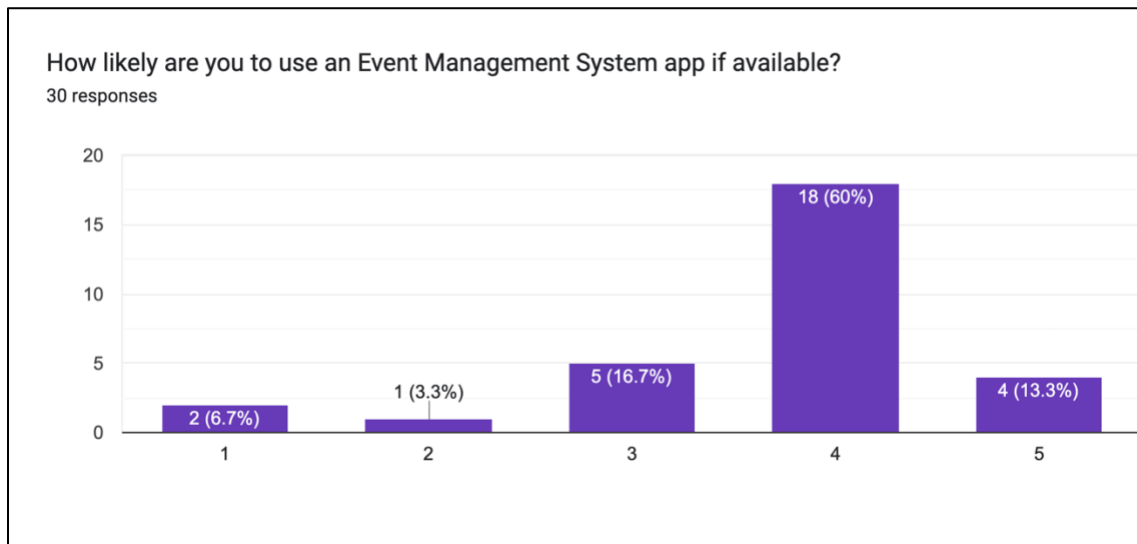


Figure 3.3: Likelihood of Respondents to Use EMS

This shows that there is a high demand for an EMS in the university at least among students and committee members. The author did not receive any responses from Event organizers for the survey so any opinions expressed by the responses must not be taken to include their views. The author did however manage to interview one Event organizer which will be touched on in a later section.

## Section B: For Students



Students widely differ on how they find out about events on campus. Of the 27 students surveyed, 33.3% of them do so through Instagram, while 18.5% learn about events through WhatsApp. This means a majority of students discover events through platforms independent of MMU. This shows the significant role played by social media platforms in spreading word regarding an event. This is especially clear when compared with the percentage of students who find out about events through the official MMU portal, numbering at only 14.8%, being only a little more common than Posters (11.1%). Also significant is the role played by friends as seen from how 22.2% of students find out about their events through their acquaintances as seen in Figure 3.4.

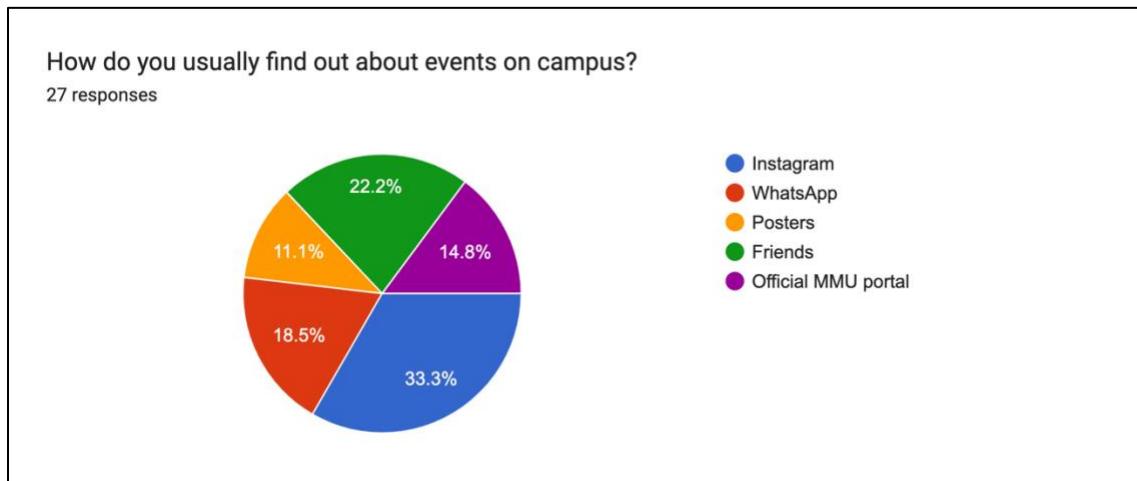


Figure 3.4: Most Common Channels for students to Learn about Events

Mirroring the diversity of channels used for finding out about events, students also widely differ in their perception of the difficulty of registering for them. Among the 27 student respondents, only two answered that it was difficult while nine answered that it was easy. The remaining 15 respondents answered that it was neither as seen in Figure 3.5 where a rating of 1 stands for “Very difficult” while a score of 5 stands for “Very easy”.

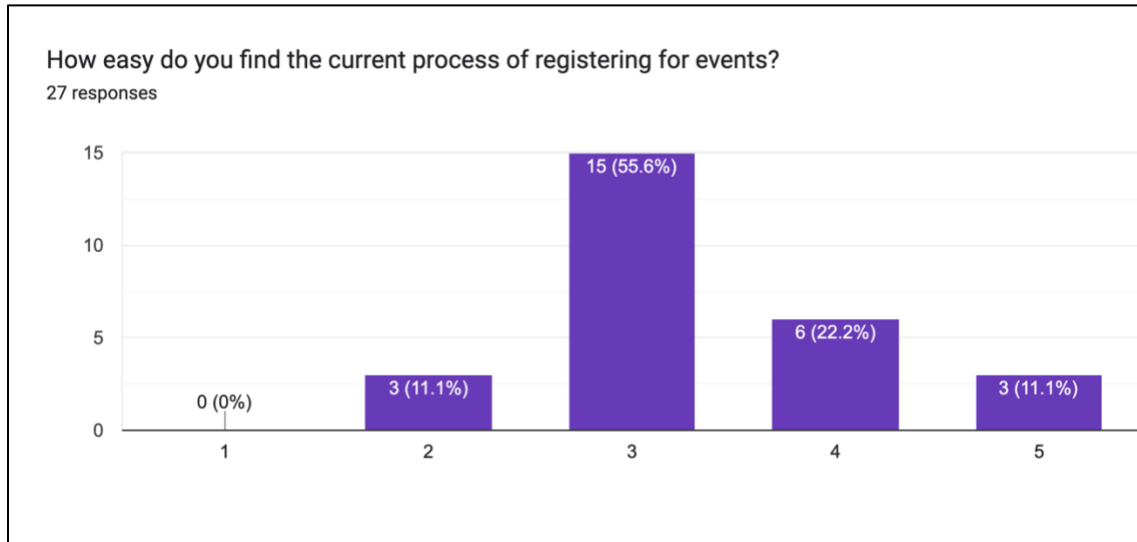


Figure 3.5: Difficulty of Event Registration

While only 2 respondents were of the opinion that event registration was difficult, it is apparent that it has its problems. Of the 27 students who responded to the survey, 16 thought the event registration process involved confusing instructions and was hard to track, 12 pointed out that no reminders are given upon registration, 11 answered that response or confirmation in regards to event registration are slow, and 9 pointed to the non-centralization of event registration links as seen in Figure 3.6.



Figure 3.6: Problems with Event Management Processes

In terms of what actually causes students to register for events in the first place, the most common source of influence seems to be Friends, with 22 (81.5%) respondents answering that Friends Joining are what influences them to

participate in an event. Other than that, the most common sources of influence for students to participate in an event are certificates (55.6%), free gifts (44.4%), location (44.4%), i-Cadet requirements (29.6%), the event's relevance to studies (25.9%) and the event's being organized by a specific club (14.8%). This indicates that there are a variety of reasons for students to join events as seen in Figure 3.7.

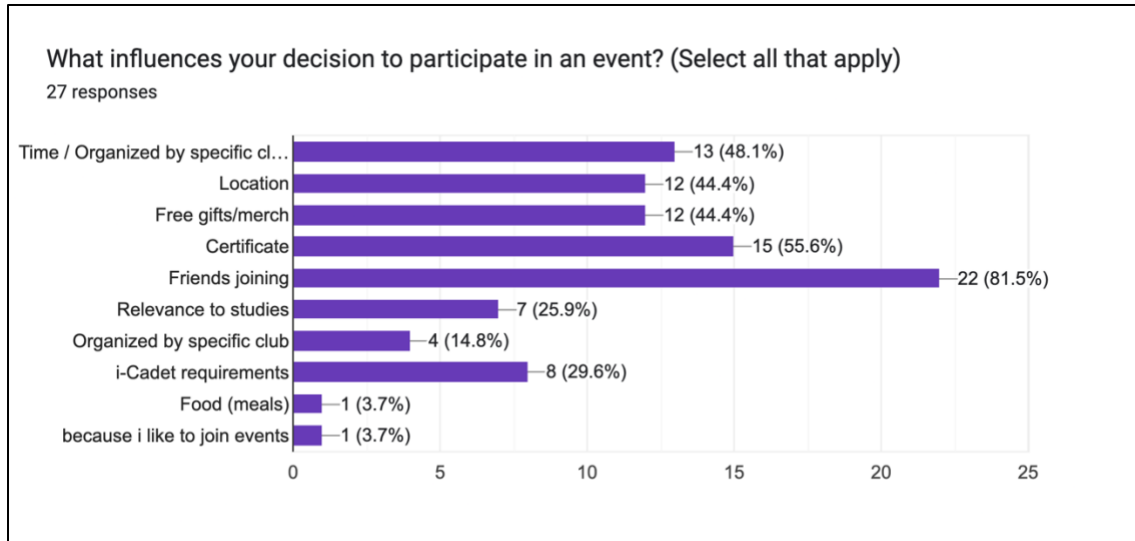


Figure 3.7: Respondents' Sources of Influence for Joining Events

Among the problems faced by students about campus events, the far most common would be scheduling clashes, with 24 (88.9%) students reporting it as one of the problems they face. This shows that there is a large overlap between student classes and university events. Second to it is lack of updates regarding events, reported by 13 (48.1%) students, followed by poor communication, poor organization and hard-to-track participation, each reported by exactly 11 (40.7%) students as seen in Figure 3.8. While this suggests problems on part of committee members and event organizers, the author believes it may also be indicative of the inefficiency of current event notification channels.

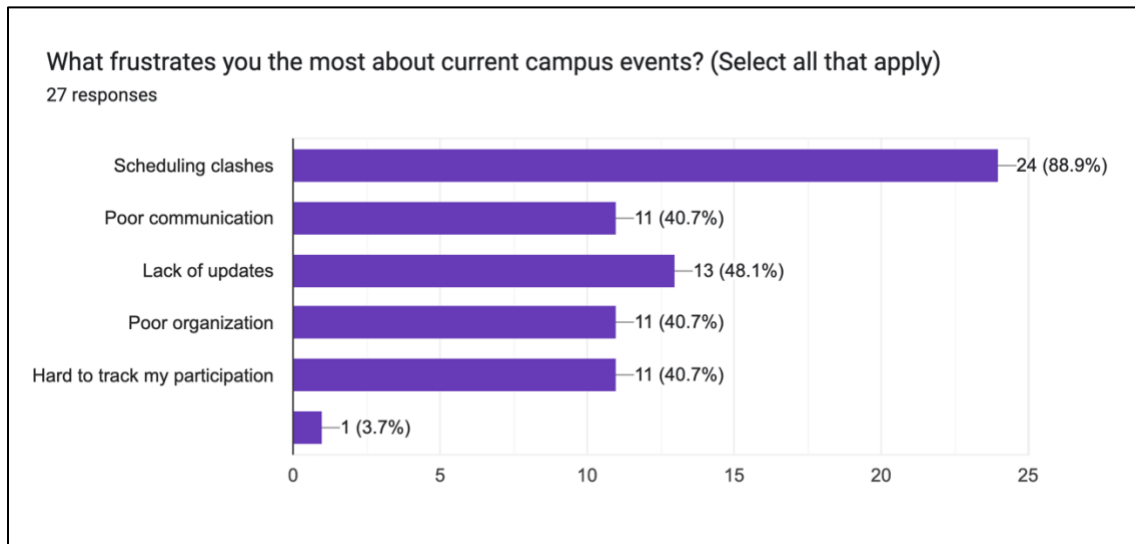


Figure 3.8: Problems Respondents Encounter with University Events

Of the several possible features given for an EMS as seen in Figure 3.9, the most popular among students was the centralization of all university events in a single location, selected by 24 (88.9%) students. Following that is registration and e-ticket event reminders, selected by 22 (81.5%) students. Other features selected by a majority of students are event calendar synchronization with personal timetable (19 or 70.4% of students), event history and certificates (18 or 66.7% of students), and push notifications (15 or 55.6% of students). Of notable popularity, although not reaching a majority, are smart recommendations (13 or 48.1% of students) and feedback systems (11 or 40.7% of students).

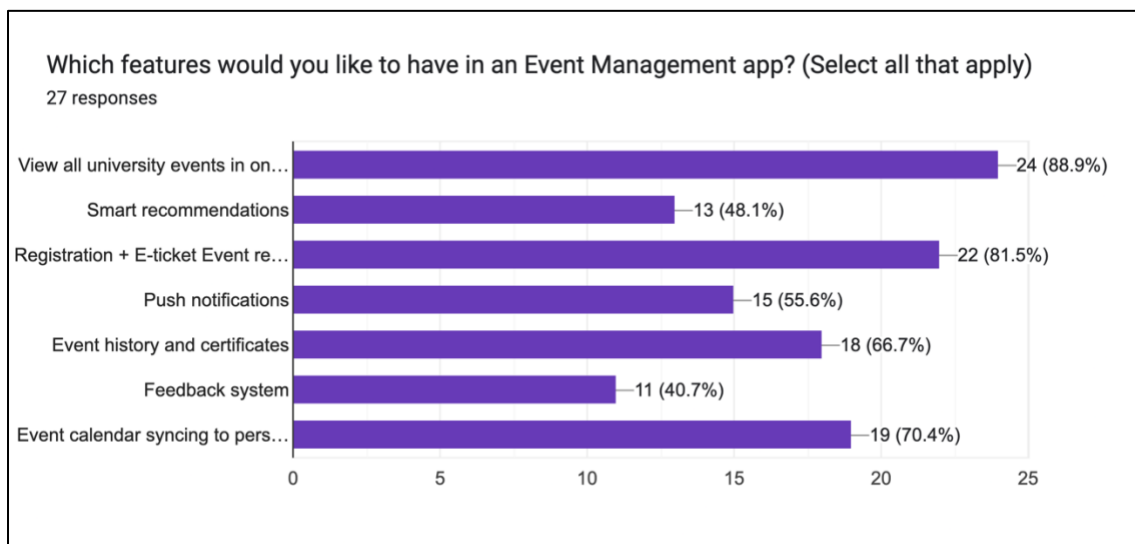


Figure 3.9: Respondents' Desired Features for an EMS

## Section C: For committee members

Event planning among committee members is done through various means. Of the various options given, each was selected by one committee member, as shown in Figure 3.10. For this reason, it is difficult to accurately assess current Event Management tools as no Event Management tool is used by more than one committee member.

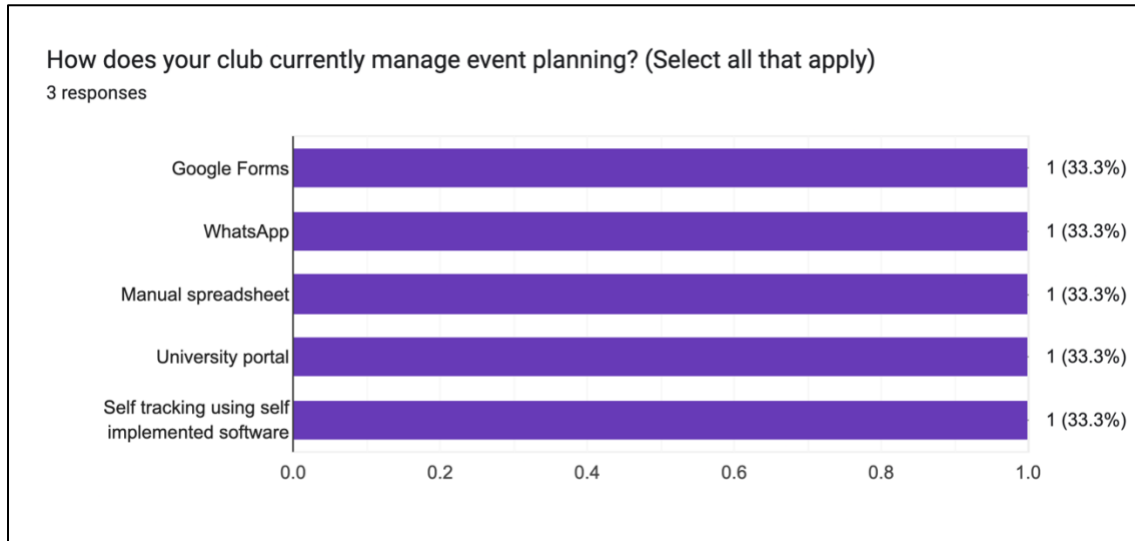


Figure 3.10: Committee members' Preferred Event Management Tools

Committee members all differed on how effective their current event management workflow. One rated theirs as very ineffective, one as neither ineffective nor effective, and one as very effective, shown in Figure 3.11 where a score of 1 stands for "Very ineffective" and a score of 5 stands for "Very effective". This shows that the current workflows of committee members are very different.

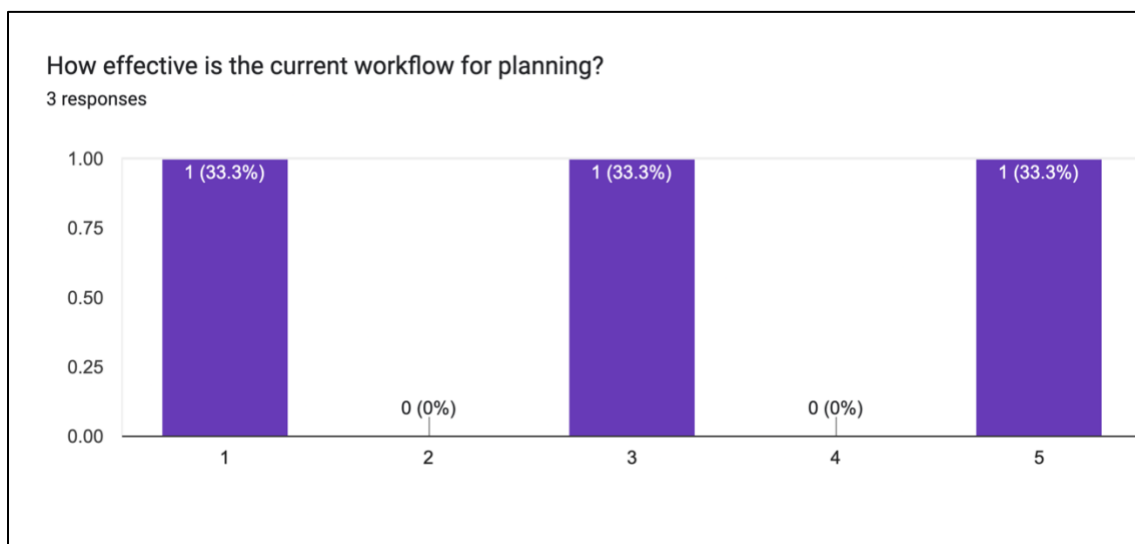


Figure 3.11: Effectiveness of committee members' Workflow

Of the numerous potential challenges when planning an event given as options for the survey as in Figure 3.12, they all seem to occur to some extent. As the sample size of committee members is small, it is difficult to tell how common these challenges are within MMU committees. Nonetheless, they present existing problems with current event management processes in MMU.



Figure 3.12: Challenges of Committee Event Management

Of the possible options for tools that committee members would like to have to help with event management, a “venue availability” tool that checks whether a venue is available was selected by all respondents. This points to venue availability being a main concern among event planners. Following that with two votes are a centralized event dashboard, a checklist/task management, and automated participant attendance, followed by all the other options with only one vote as seen in Figure 3.13.

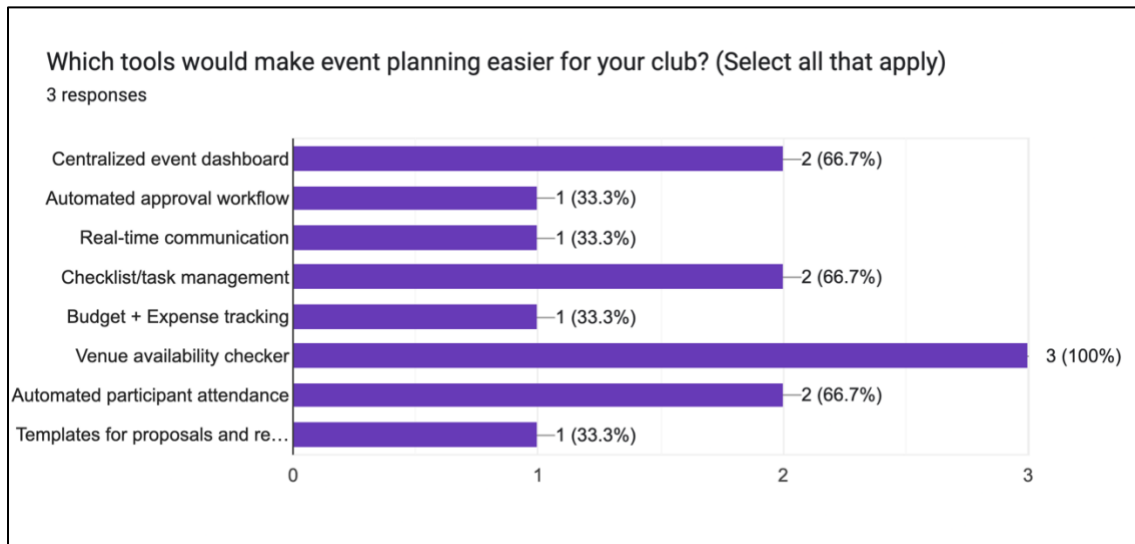


Figure 3.13: Tools that would Make Event Planning Easier

The period of event planning until final approval seems to take three weeks according to the responses of two respondents as in Figure 3.14. Another respondent answered with a larger value but without the time unit specified, making it unclear whether they intended to answer in days or weeks.



Figure 3.14: Period of Event Planning for committee members

On a semantic scale where a score of 1 stood for “Very problematic” and a score of 5 stood for “Very smooth”, one respondents responded that the process of venue booking was the former, while the two other respondents responded that it was neither as in Figure 3.14. This suggests that venue booking, although not a problem for all committee members, perhaps could be improved upon in some aspects.

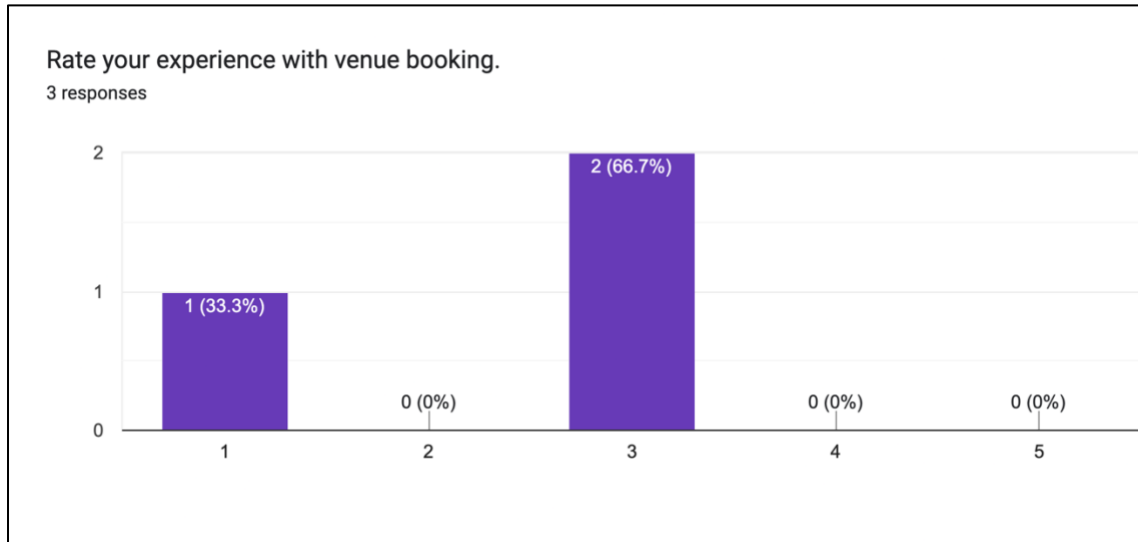


Figure 3.15: Smoothness of Venue Booking Process

None of the respondents seem to face difficulties in acquiring approval for events. This shows that the current event approval process is not causing any problems for committee members as in Figure 3.15.

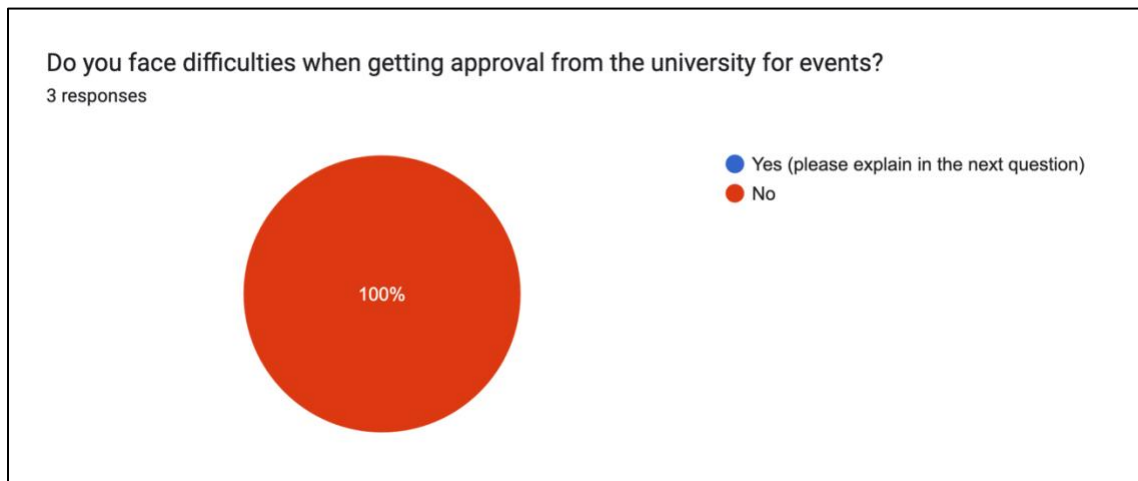


Figure 3.16: Frequency of Difficulties When Acquiring Approval for Event

## Section D: For Event organizers

The author did not receive any responses for the survey from Event organizers.

### 3.2.3 Interview

## Section A: For students

### Subsection A: Experience and Behaviour



**Question 1:** Can you walk through the last campus event you attended, from discovering it to attending it?

**Question objective:** To inquire the last campus event attended by the student.

**Question 2:** How did you register, and how long did it take?

**Question objective:** To inquire how the registration process of a student for an event unfolds.

**Question 3:** What tools or platforms were involved?

**Question objective:** To inquire what tools or platforms are used for registration in campus events.

**Question 4:** Was there any part of the process that you found annoying?

**Question objective:** To inquire what parts of the registration process that students find annoying.

### **Subsection B: Pain Points and Workarounds**

**Question 5:** What problems do you usually face when joining campus events?

**Question objective:** To elicit problems usually encountered by students when registering for a campus event.

**Question 6:** Have you ever missed an event you wanted to attend? Why?

**Question objective:** To elicit reasons students may miss an event.

**Question 7:** When things go wrong (no confirmation, confusion), how do you usually handle it?

**Question objective:** To elicit methods students employ to handle problems that arise when attempting to join an event.

### **Subsection C: Scheduling and Clashes**

**Question 8:** Have you experienced event clashes with classes or other events?

**Question objective:** To elicit how common it is for event overlaps with classes to occur among students.

**Questions 9:** How do you currently check if an event clashes with your timetable?

**Question objective:** To elicit ways in which students check schedule overlaps between events and classes.

### **Subsection D: Communication and Engagement**

**Question 10:** How do organizers usually communicate changes or updates?

**Question objective:** To elicit ways organizers communicate changes or updates to students.

**Question 11:** Do you feel informed enough before and during events?

**Question objective:** To elicit the effectiveness of the ways organizers communicate with event participants for important notifications.

### **Subsection E: Ideal System**

**Question 12:** If you could design the “perfect event app” for students, what must it have? Why?

**Question objective:** To elicit the features essential for a ‘perfect event app’ in the eyes of the students.

**Question 13:** What would make you actually use such a system instead of WhatsApp/Instagram?

**Question objective:** To elicit what it would take for students to switch to an EMS instead of relying solely on other mediums such as WhatsApp and Instagram.

**Question 14:** Would you trust the system for official certificates or attendance tracking?

**Question objective:** To elicit the level of trust an EMS would have in the view of students.

### **Subsection F: Closing**

**Question 15:** What would make you stop using an Event Management System even if it exists?

**Question objective:** To elicit features or design decisions to avoid for an EMS.

## **Section B: For Committee Members**

### **Subsection A: Workflow Understanding**

**Question 1:** Can you describe the full process of organizing an event in your committee?

**Question objective:** To elicit the process of organizing an event by committee members.

**Question 2:** Which tools do you use at each stage? (Planning, approval, promotion, execution, reporting)

**Question objective:** To elicit the tools used by committee members in each stage of the event organization process.

### **Subsection B: Bottlenecks and Pain**

**Question 3:** Which part of event planning is the most time-consuming?

**Question objective:** To elicit the most time-consuming part of event planning.

**Question 4:** What causes delays most often?

**Question objective:** To elicit the causes of delays of events by committee members.

**Question 5:** Have you ever had to redo work because of miscommunication or lost information?

**Question objective:** To elicit whether miscommunication or information loss has led to parts of the event planning process having to be redone.

### **Subsection C: Approval and Coordination**

**Question 6:** How do you submit proposals and get approvals?

**Question objective:** To elicit how proposals are submitted and how approvals are acquired.

**Question 7:** How transparent is the approval process?

**Question objective:** To elicit the transparency of the process of acquiring an approval for an event.

**Question 8:** Have you experienced last-minute rejection or venue issues?

**Questions objective:** To elicit how common it is for events to be abruptly rejected or interrupted at the last minute.

### **Subsection D: Attendance and Reporting**

**Question 9:** How do you track attendance during events?

**Question objective:** To elicit how committee members track attendance.

**Question 10:** How reliable is this method?

**Question objective:** To elicit the reliability of the methods committee members use to track attendance.

**Question 11:** What reports are you required to submit after events?

**Question objective:** To elicit what reports have to be submitted and to whom after events are finished.

### **Subsection E: System Adoption**

**Question 12:** If an Event Management System replaced your current tools, what would worry you?

**Question objective:** To elicit the worries committee members may have if an EMS were available.

**Question 13:** What features would make your committee insist on using it?

**Question objective:** To elicit features essential for an EMS.

**Question 14:** Would automation help or complicate your work?

**Question objective:** To elicit the effects automatic scheduling would have on event planning.

### **Subsection F: Closing**

**Question 15:** If you could remove one step from event planning, what would it be?

**Question objective:** To elicit what steps of event planning committee members would want to be removed or simplified.

## **Section C: For organizers**

### **Subsection A: Policy and Control**

**Question 1:** Can you explain how event approvals are handled in your department?

**Question objective:** To elicit how event approvals are handled by organizers.

**Question 2:** What rules must committee comply with before approval?

**Question objective:** To elicit rules committees must comply with to acquire event approvals.

### **Subsection B: Volume and Load**

**Question 3:** How many event requests do you manage in a typical semester?

**Question objective:** To elicit the number of event requests organizers typically receive in a single semester.

**Question 4:** What becomes difficult when volume increases?

**Question objective:** To elicit the problems that come with an increase in the volume of event requests.

### **Subsection C: Visibility and Monitoring**

**Question 5:** How do you track:

- Event status?
- Venue usage?
- Attendance?

**Question objective:** To elicit how organizers track event status, venue usage and attendance.

**Question 6:** What information do you wish you had instantly but don't?

**Question objective:** To elicit what information regarding events do organizers find important.

### **Subsection D: Risk and Compliance**

**Question 7:** What risks worry you most in student-organized events?

**Question objective:** To elicit the risks involved in student-organized events in the view of organizers.

**Question 8:** How do you ensure compliance with university policies?

**Question objective:** To elicit how organizers ensure events are compliant with university policies.

### **Subsection E: System Weaknesses**

**Question 9:** What frustrates you most about the current event management process?

**Question objective:** To elicit frustrations organizers have with current event management processes.

**Question 10:** Where do challenges usually happen?

**Question objective:** To elicit where challenges usually occur in the event management process.

### **Subsection F: Decision-Making**

**Question 11:** What would convince you to adopt a new Event Management System?

**Question objective:** To elicit what it would take for organizers to adopt an EMS.

**Question 12:** What would block its implementation?

**Question objective:** To elicit what could cause the implementation of an EMS to be blocked.

### **Subsection G: Future Vision**

**Question 13:** Ideally, how should event management work in 3–5 years?

**Question objective:** To elicit the vision organizers have for how event management should be done in the near future.

**Question 14:** How could data or automation help your department?

**Question objective:** To elicit the effects of automated scheduling on event management in the view of organizers.

### *3.2.3.1 Analysis on Interview Results*

Findings of the interviews show numerous issues with current MMU event management processes. From the perspective of students, the biggest one is that there no centralized place to search for events. To add to this, to look for events, one has to go through multiple fragmented platforms such social media platforms such as Instagram or the MMU portal. This puts a heavy burden on students just to know about the university events. It also makes it very likely for the students to either not know about them at all or to only know when it is too late and the events are finished.

Registration is also seen as something that is unreliable. Students reported having to repeatedly sign a never-ending number of forms to register, while also not being notified if their registration was ever accepted. This again is due to the fragmentation in the channels used by committees to inform students of their registration confirmation. Even if the student is checking the proper channels, they may be lost in the sea of emails and messages that they have to go through as part of the natural course of being a university student. This causes students, even those with genuine interest in an event, to not join, simply because of the lack of clarity as to whether they have actually registered for the event.

Also seen as a major issue was the difficulty of ensuring no schedule clashes occur between events and classes and other things. Many students reported that they had to do so manually which is highly prone to oversight. When a schedule clash does occur, some students are also told that they would be receive a class exemption notice, enabling them to participate in the event without worrying about their class attendance. But this also was unclear a lot of the time, reducing students' trust in the event management processes and making them more likely to not attend university events.

From the perspective of organizers, the main challenge and problem was in event approval which involves estimating the attendance of the proposed event and allocating the appropriate venue, both of which can be difficult. If done improperly, it will result in a waste of university resources. There is also the risk of an approved event, possibly causing some form of injury or hurt, which requires organizers to also ensure to lessen an event of such kind from occurring.

## **3.3 Requirements**

From the findings of the interviews and survey conducted, the author was able to derive the requirements for the EMS. The requirements are divided into three which are functional requirements, non-functional requirements and user requirements.

### **3.3.1 Functional Requirements**

### **3.3.1.1 Login Function**

This login function must work in a way that allow users to key in their login credentials. After that, users would be able to click the login button to their account and hence, being able to enter the main menu of the EMS and able to use main functions of the system based on their user role. If the user is not yet registered to the system, they could click the register button to register for a new account with their details. After that, the user could enter their user details and new login credentials and enter the main menu of the system which will differ depending on the user's user role.

### **3.3.1.2 Event Discovery Function**

The event discovery function must work in a way where students are able to look up events by either searching the name of the event or by filtering events by event type, date or committee. If neither the search nor the filter tools are used, then all the events will be arranged in the list in chronological order from earliest event to latest event.

### **3.3.1.3 Registration Function**

The EMS shall allow students to register for events that they choose. Upon registration, they will immediately be given registration confirmation in the form of a notification. When a registration is confirmed, this event will be added to a "My Events" list which contains all the events that the students have registered for. If they cancel the event, the event will be removed from the list and a notification for event cancellation confirmation will be given.

### **3.3.1.4 Notification Function**

The notification function must work in a way where students are given a notification upon registering for an event. Students must also be given a notification if the events they have registered for have received any updates to its event details. When the registered event is imminent, students will receive a notification reminder a week, day, and hour before it starts. For notifications, the EMS will support both push and in-app notifications. All notifications will then be stored in a notifications list for the user to see all the notifications that they have received.

### **3.3.1.5 Scheduling Function**

The scheduling function must work in a way where events that students have registered for will be automatically added to an in-app calendar. The student may also add any events of their own to the calendar such as classes or tests to ensure no event clashes occur.

### **3.3.1.6 Attendance Function**



The EMS will support an attendance recording mechanism whereby committee members will be able to generate a QR code on site for event attendees to scan to put in their details to confirm attendance. After they have taken their attendance, the attendance will be stored attendance history where students will be able to view all their past attendances.

#### **3.3.1.7 Venue Management Function**

The EMS will provide a venue availability checker where committee members and administrators will be able to check if a venue is available at a certain time and day. If the venue is available, committee members may request for venue at the set time after which an administrator will have to approve or reject the request. If the venue is not available, then committee members will be informed that the is taken at that time. This will prevent venue double-booking from occurring.

#### **3.3.1.8 Event Planning Function**

The EMS will include an organizer dashboard which will assist event organizers in managing event drafts, attendance, and registration. Alongside that, the EMS will also provide a task management checklist which event organizers can use to track event preparation tasks.

#### **3.3.1.9 Event Analytics Function**

Upon completion of an event, the EMS will generate basic analytics for admins and organizers which will include figures such as event registration and attendance numbers. For each venue, the EMS will also provide utilization reporting which will show how efficiently they are used.

### **3.3.2 Non-Functional Requirements**

#### **3.3.2.1 Usability**

The EMS will be easy to learn for students and committee members where tasks such as registration and attendance will take a minimal number of steps. Upon completion of registration or taking attendance, the EMS will provide immediate status feedback.

#### **3.3.2.2 Performance**

The EMS will return browsing and search results under 2 seconds under normal campus load. On the other hand, notifications will be sent within 10 seconds of time-sensitive reminders or updates.

#### **3.3.2.3 Reliability**

The EMS will provide transaction-safe registration operations to avoid the occurrence of duplicate or inconsistent statuses.

#### **3.3.2.4 Availability**

The EMS will be up 99% of the semester period.

#### **3.3.2.5 Security**

The EMS will enforce role-based access control where users are only able to parts of the system as designated by their role. Other than that, the EMS will ensure the safety of personal data by storing only what is necessary for event registration, participation, and verification. Finally, the EMS will ensure the auditability of admins by logging every action performed by them.

#### **3.3.2.6 Maintainability**

The EMS will make use of a modular design to better isolate components, hence making it easier to detect and locate errors within the code whenever a bug occurs.

#### **3.3.2.7 Scalability**

The EMS will include a database design that will allow for easy extension of the number of events, users, and attendance records to support system growth.

#### **3.3.2.8 Compatibility**

The EMS will be available on modern web browsers.

### **3.3.3 User Requirements**

#### **3.3.3.1 Centralized Event List**

Students need a single and centralized place to view events so that they do not miss any events due to the events being scattered across multiple platforms.

#### **3.3.3.2 Standardized and Simplified Event Registration**

Students need a clear and consistent way to register for events so that they can easily understand the registration process and what their registration status is, regardless of the type of event that they are joining.

#### **3.3.3.3 Event Reminder and Update Notification**

Students need timely notifications for reminders and updates for events that they registered for so that they do not miss them in case they forget about the event or miss the notifications due to them coming through multiple fragmented channels.

#### **3.3.3.4 Event Tracking and History**

Students need access to a personal record of events that they participated in and certificates they have received so they can track their participation history and retrieve their certificates when required.

#### **3.3.3.5 Standardized Event Planning Workflow**

Event organizers need a structured and standardized workflow for them to plan events and manage events as current workflows involve several disjointed tools which make coordination and cooperation between organizers difficult.

#### **3.3.3.6 Venue Availability Checker**

Event organizers need to know whether a venue is available or not before they plan for an event in order to prevent venue clashes from occurring.

#### **3.3.3.7 Participant Attendance Tracking**

Event organizers need a reliable way to track event attendance so that attendance records can be maintained for reporting, verification, and documentation.

#### **3.3.3.8 Visibility Venue Usage and Conflict**

Administrators need clear visibility of venue usage and conflicts so that they can better manage venue allocation and reduce the need for manual checks.

#### **3.3.3.9 Standardize Event Documentation**

Administrators need consistent and standardized event documentation to support reviews, compliance checks, and auditing processes.

#### **3.3.3.10 Attendance Analytics**

Administrators need access to information regarding event attendance to evaluate the effectiveness and popularity of events and make planning decisions for future events based on data.

### **3.4 Summary**

Findings of the survey and interviews conducted show that university members of various roles face numerous problems related to event management. From that, the author was able to derive the functional, non-functional, and user requirements needed to construct the EMS. With this, it can be ensured with high certainty that the EMS will meet the needs of its stakeholders and act as the solution to the many problems encountered by them with current event management processes.

## **Chapter 4: System Design**

## 4.1 Overview

The findings of the survey and interviews conducted with relevant stakeholders outlined in the previous section show there is a high demand for an EMS among members of MMU of all university roles. In this section, all the findings are employed to derive the system design of the EMS that resolves to address the issues with current event management processes. That system design is shown in the form of various diagrams as follows.

## 4.2 Rich Picture Diagram

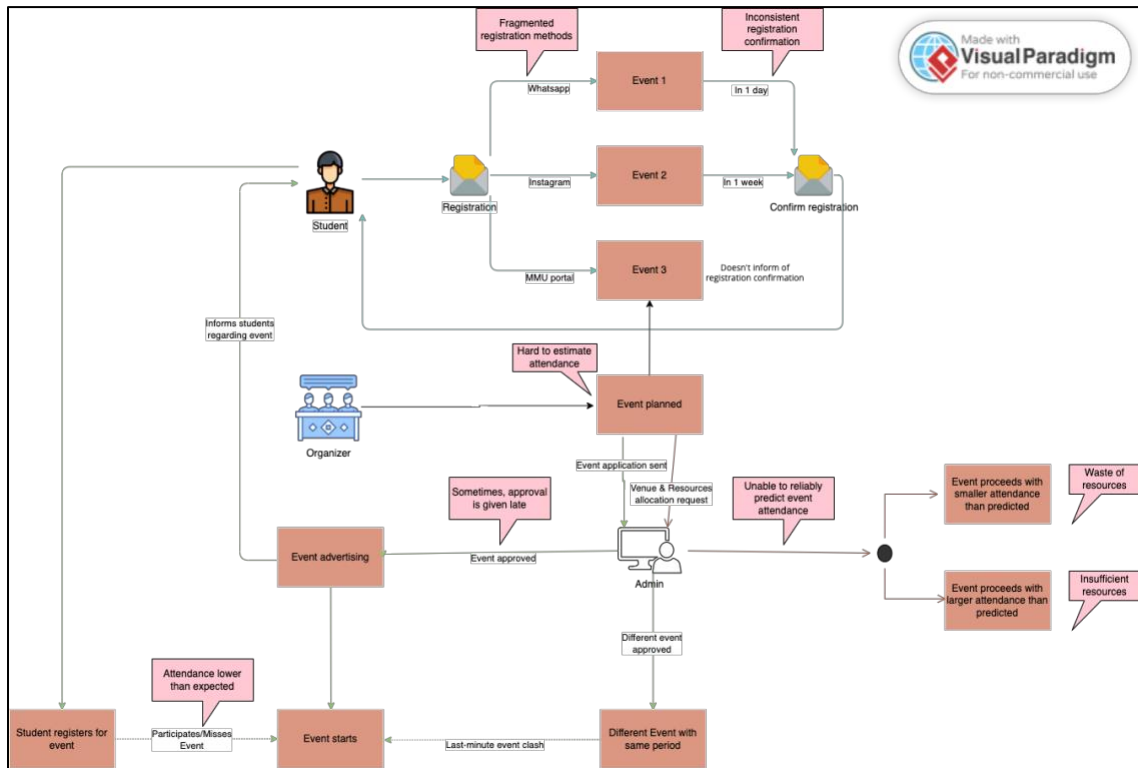


Figure 4.1: Current Event Management Process in MMU

## 4.3 Use Case Diagram

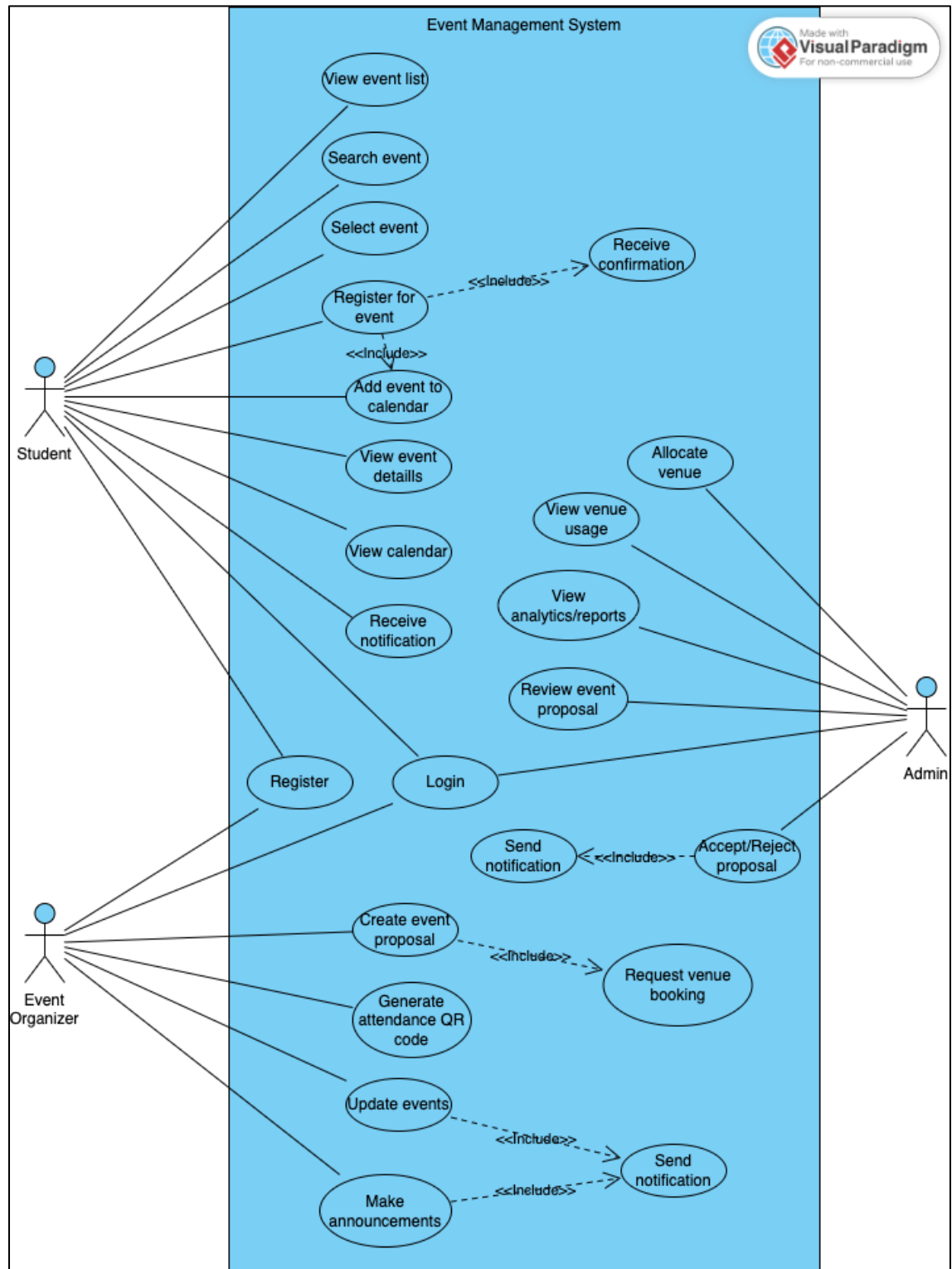


Figure 4.2: Use Case Diagram of EMS

#### 4.4 Activity Diagram

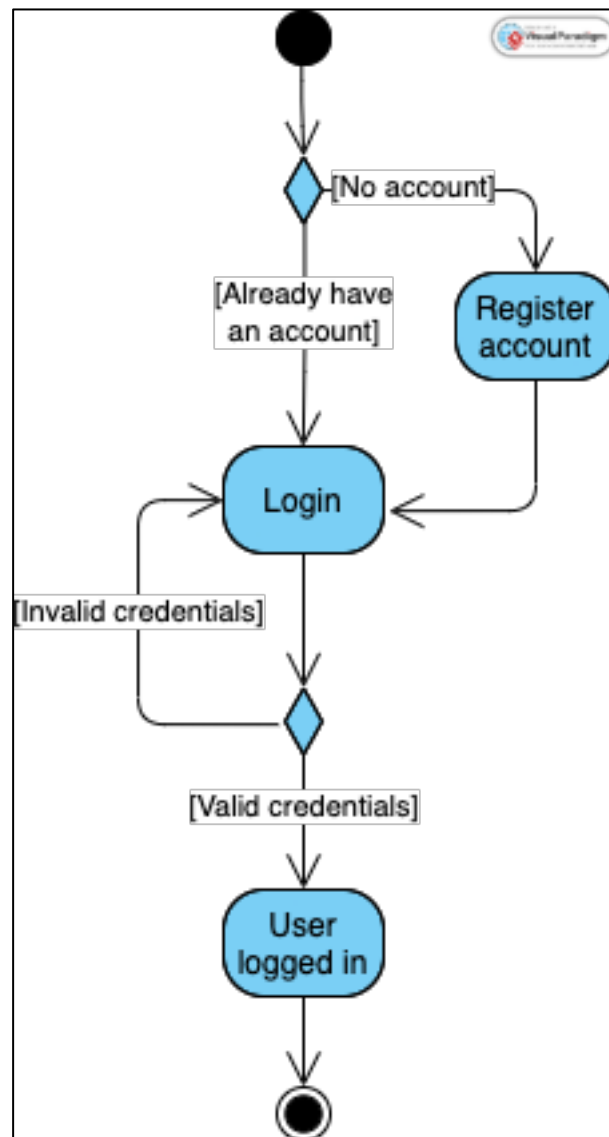


Figure 4.3: Login for Users

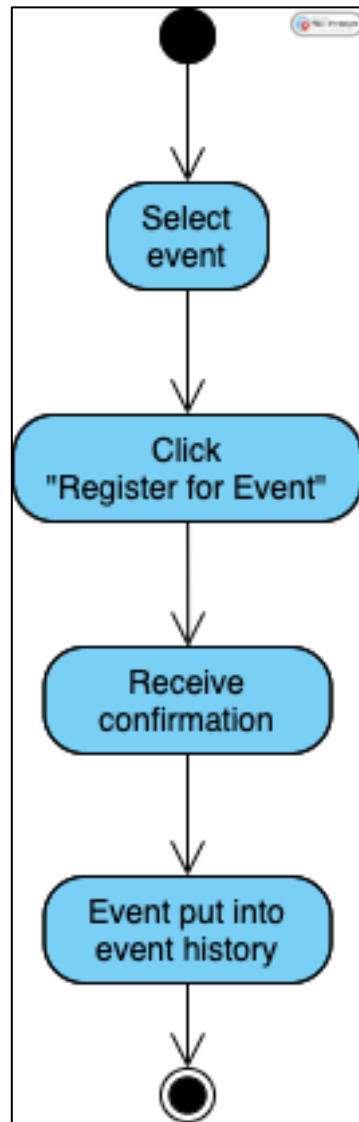


Figure 4.4: Event Registration for Students

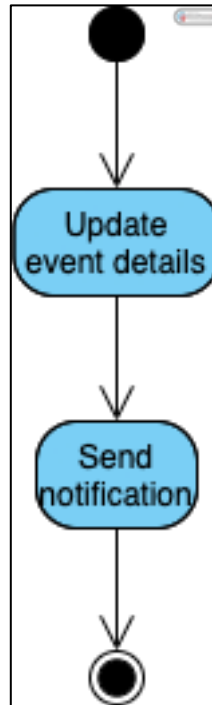


Figure 4.5: Event Organizers Updating Event Details



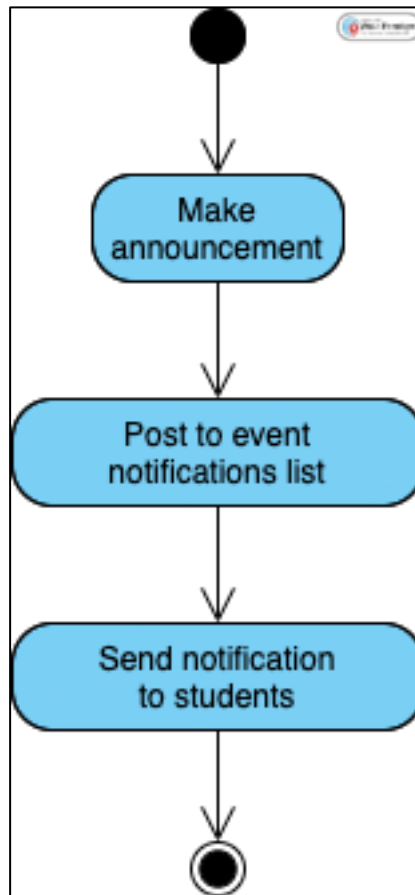


Figure 4.6: Event Organizers Making Announcements

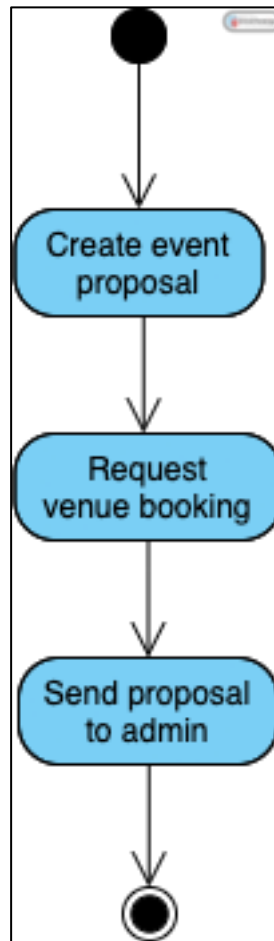


Figure 4.7: Event Organizers Creating Event Proposal

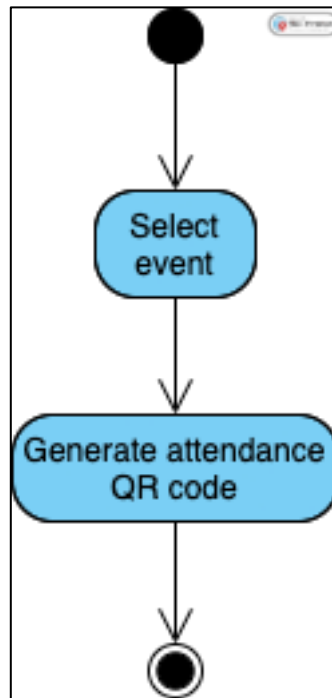


Figure 4.8: Event Organizers Generating Attendance QR Code

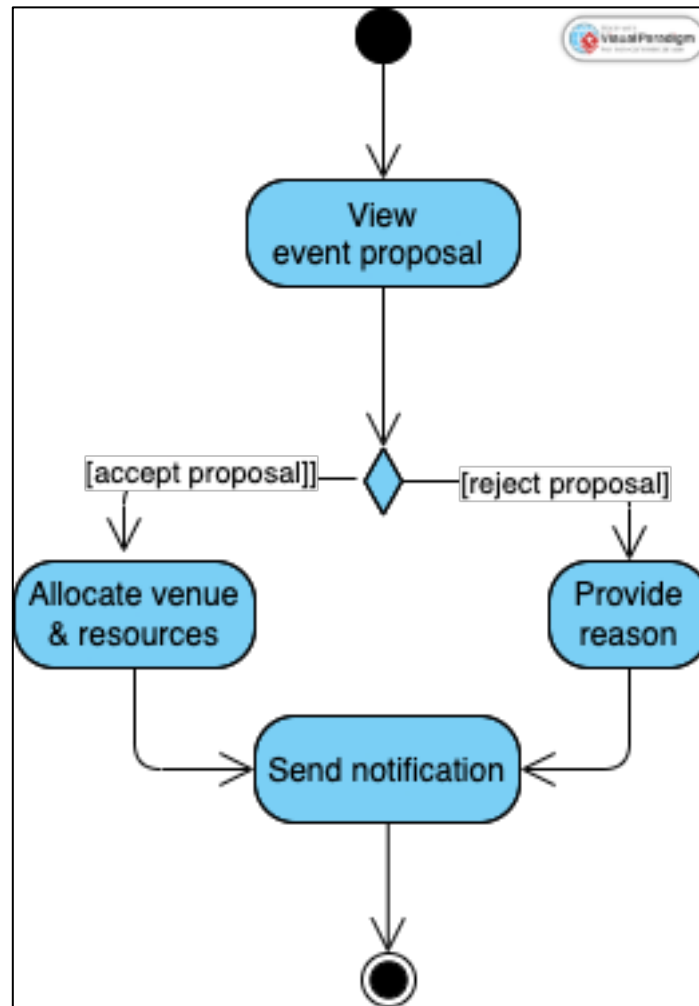


Figure 4.9: Administrators Viewing Event Proposals

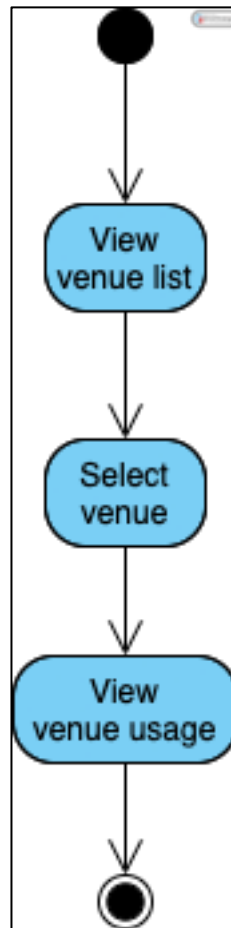


Figure 4.10: Administrators Viewing Venue Usage

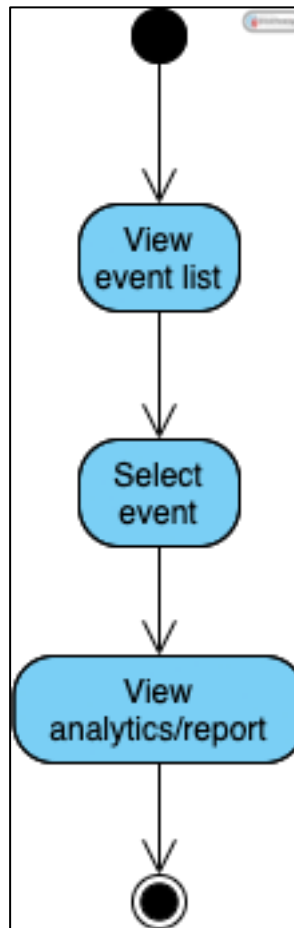


Figure 4.11: Administrators Viewing Event Analytics/Report

#### 4.5 Class Diagram

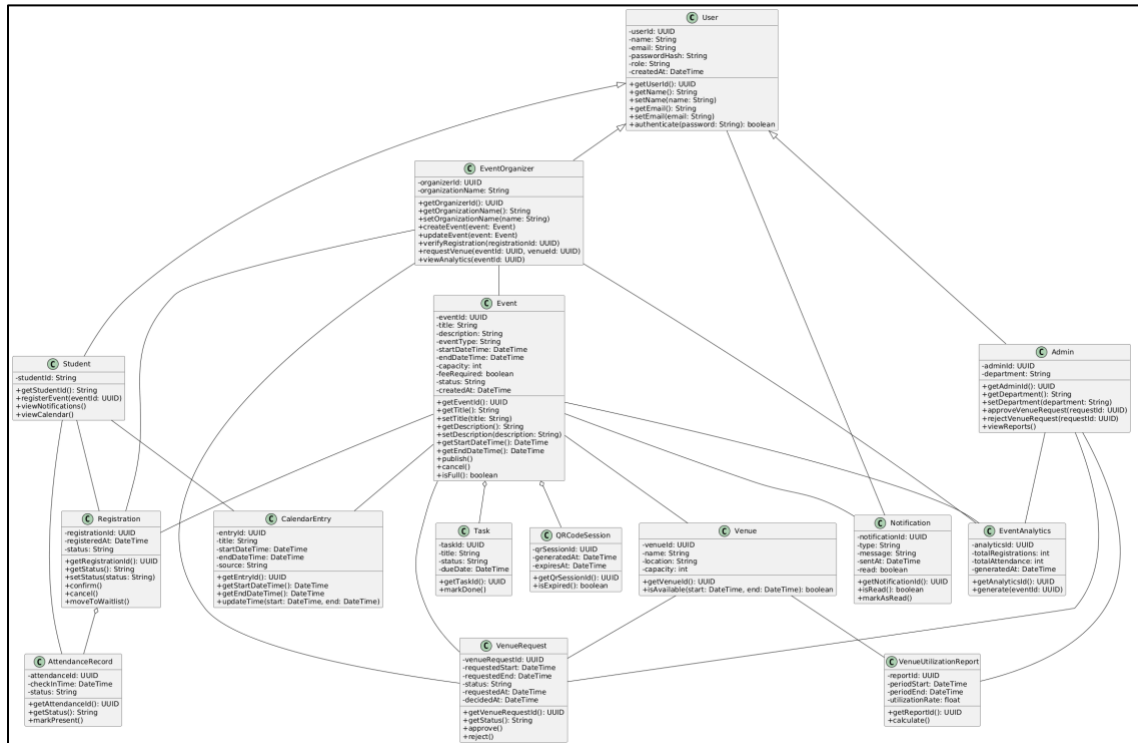


Figure 4.12: EMS Class Diagram

## 4.6 Sequence Diagram

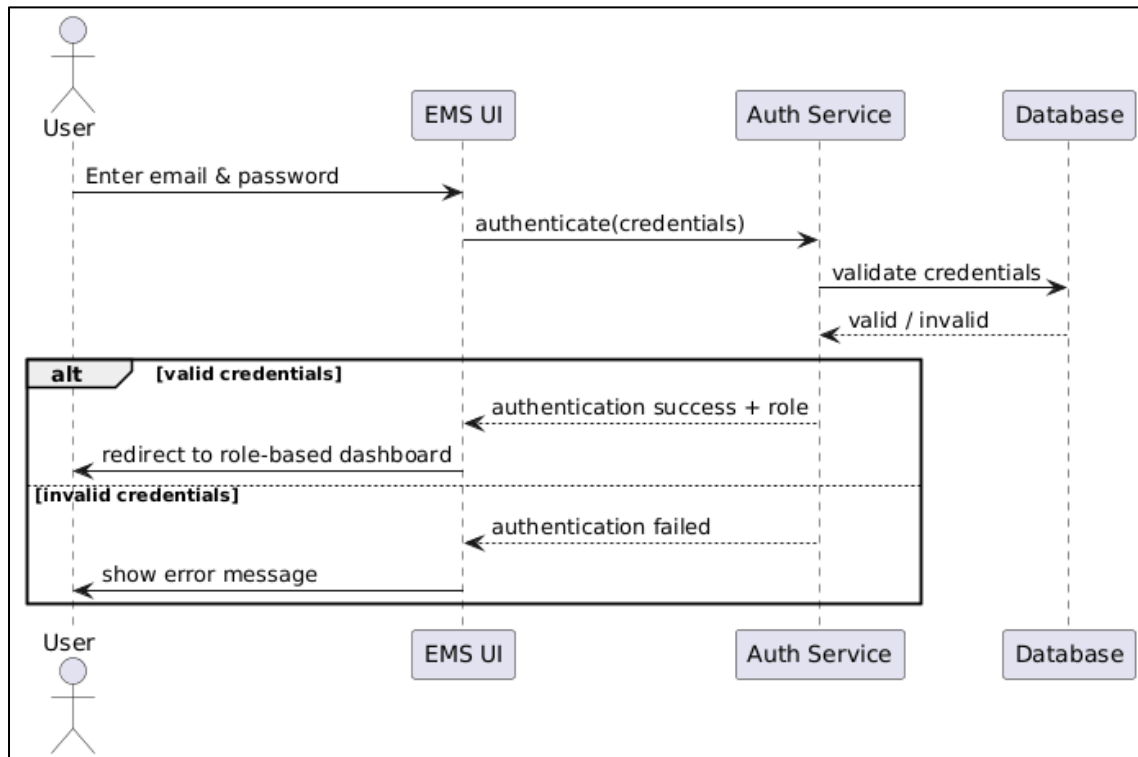


Figure 4.13: User Login Process



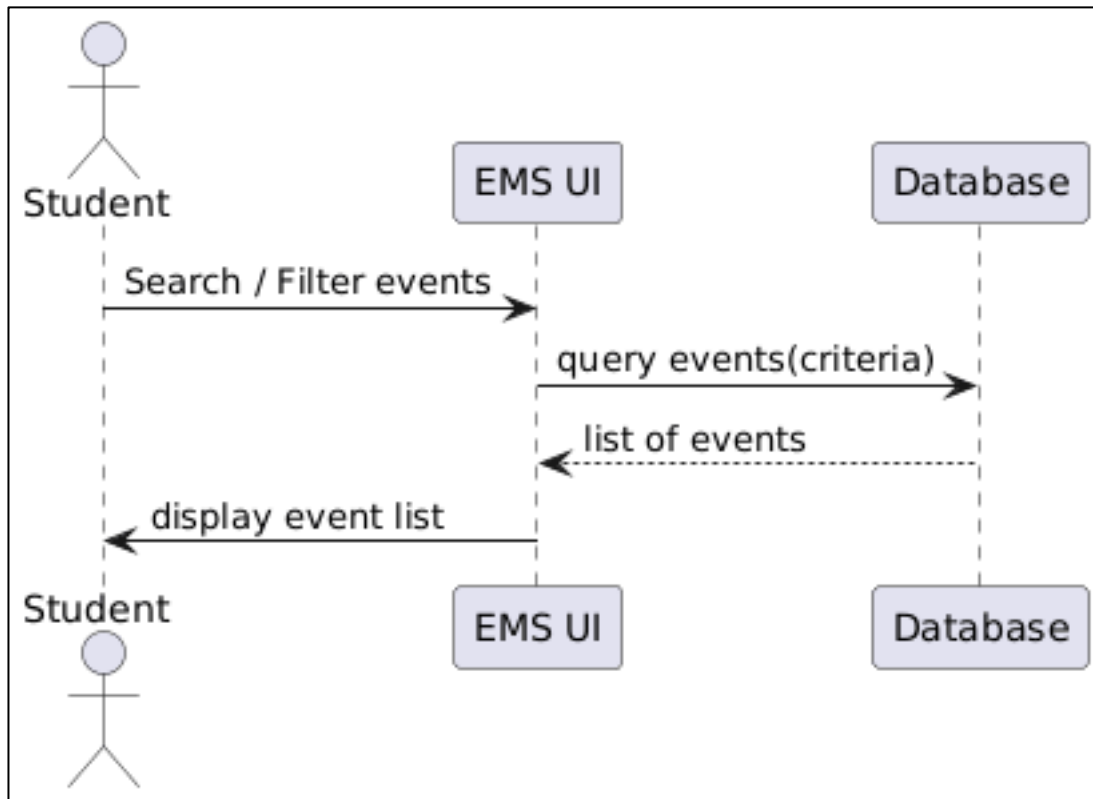


Figure 4.14: Student Event Search Process

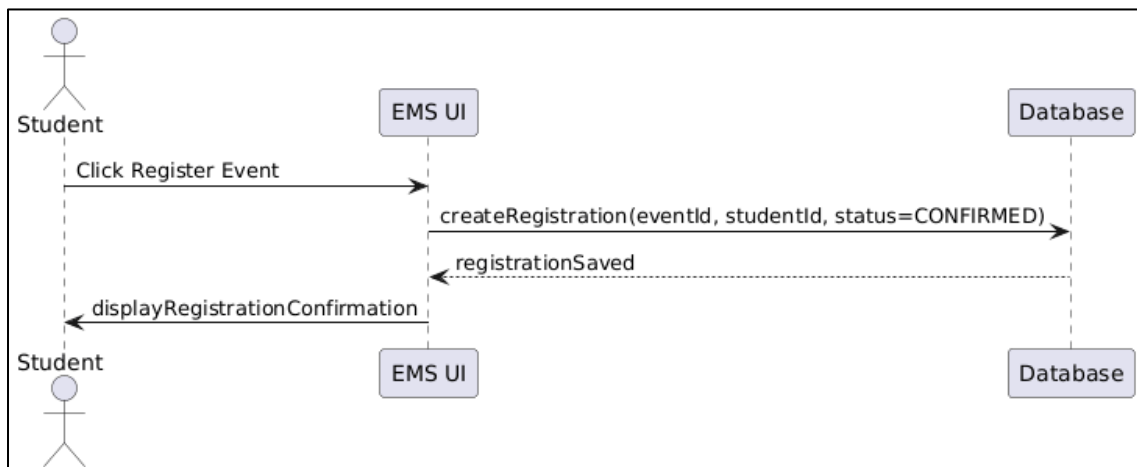


Figure 4.15: Student Event Registration Process

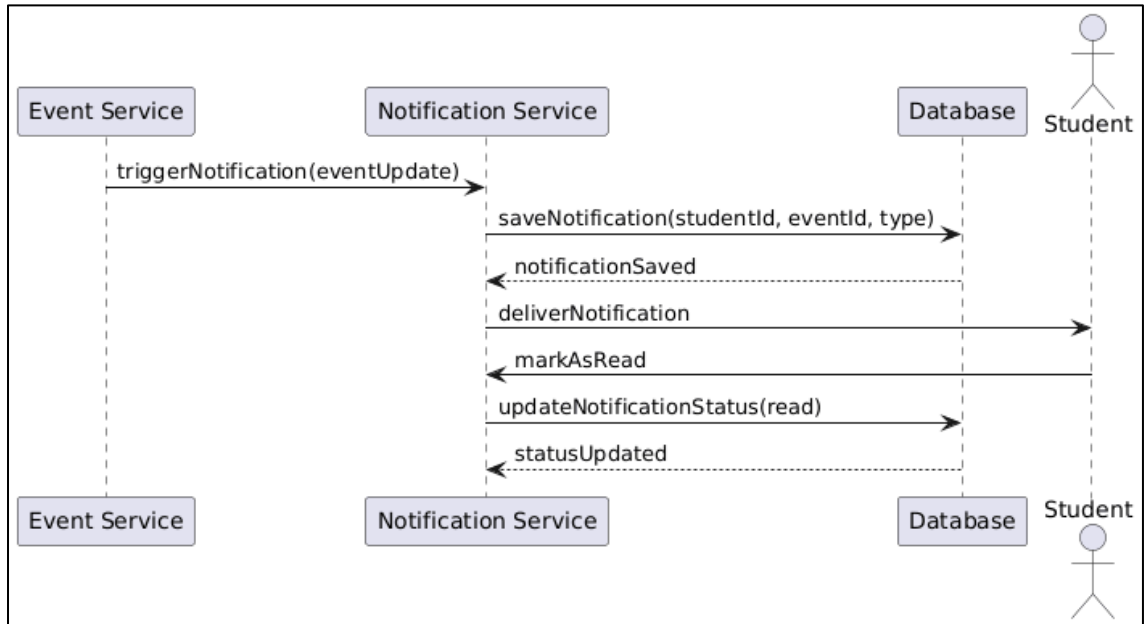


Figure 4.16: Student Notification or Reminder Flow

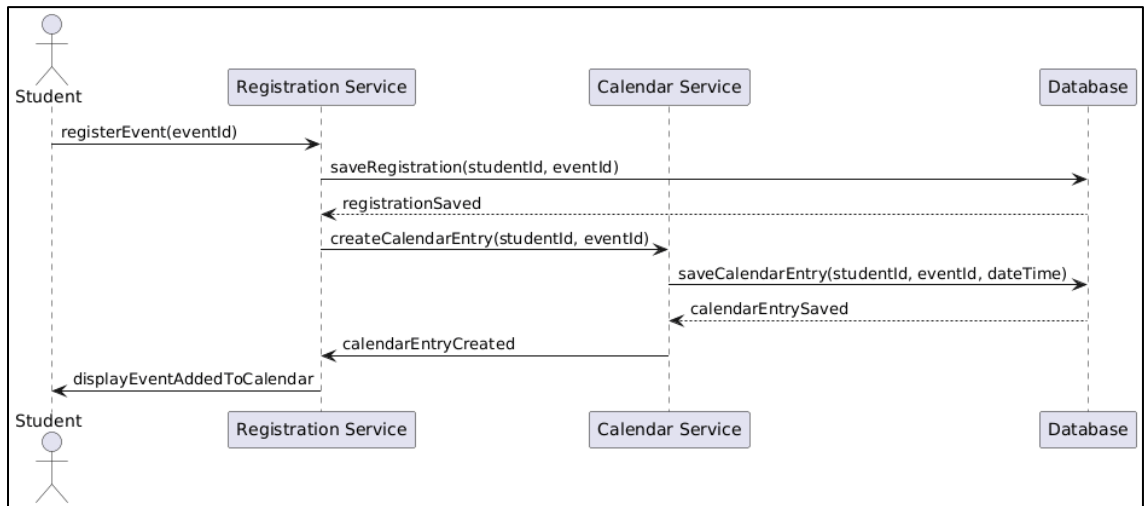


Figure 4.17: Student In-app Calendar

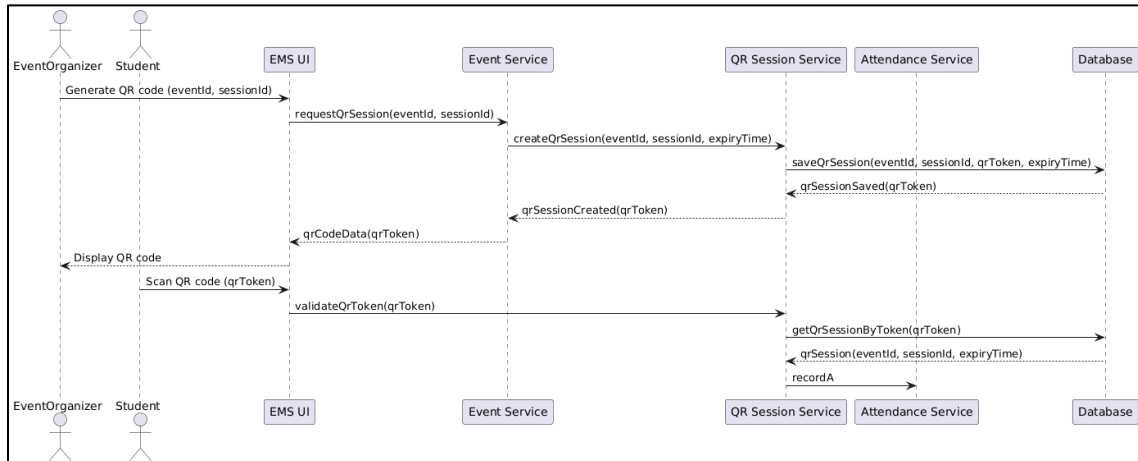


Figure 4.18: QR Code Event Attendance

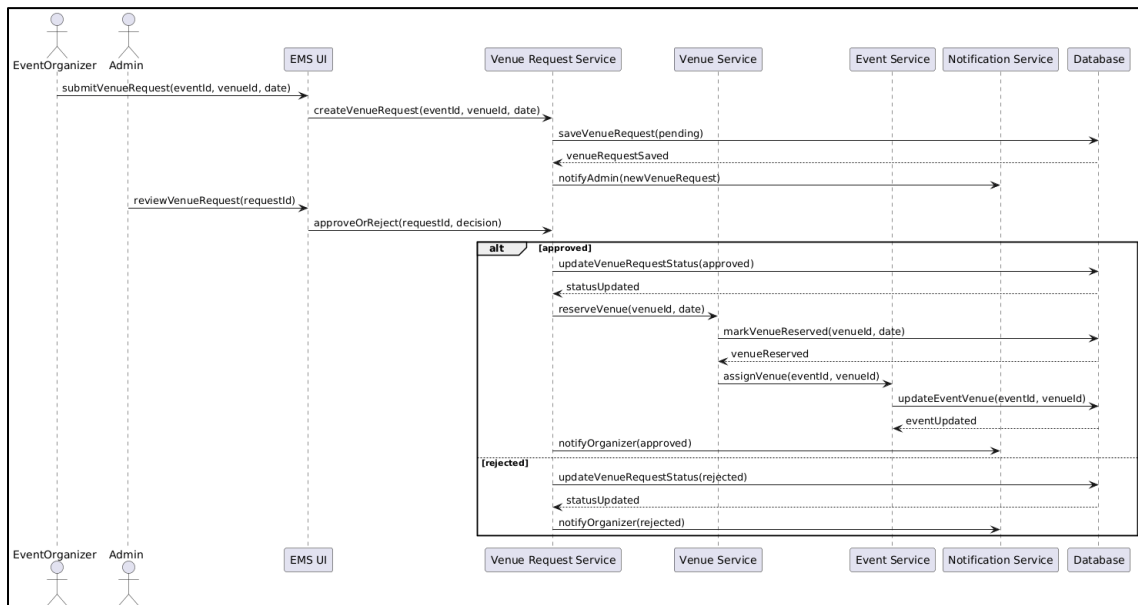


Figure 4.19: Venue Request and Admin Approval

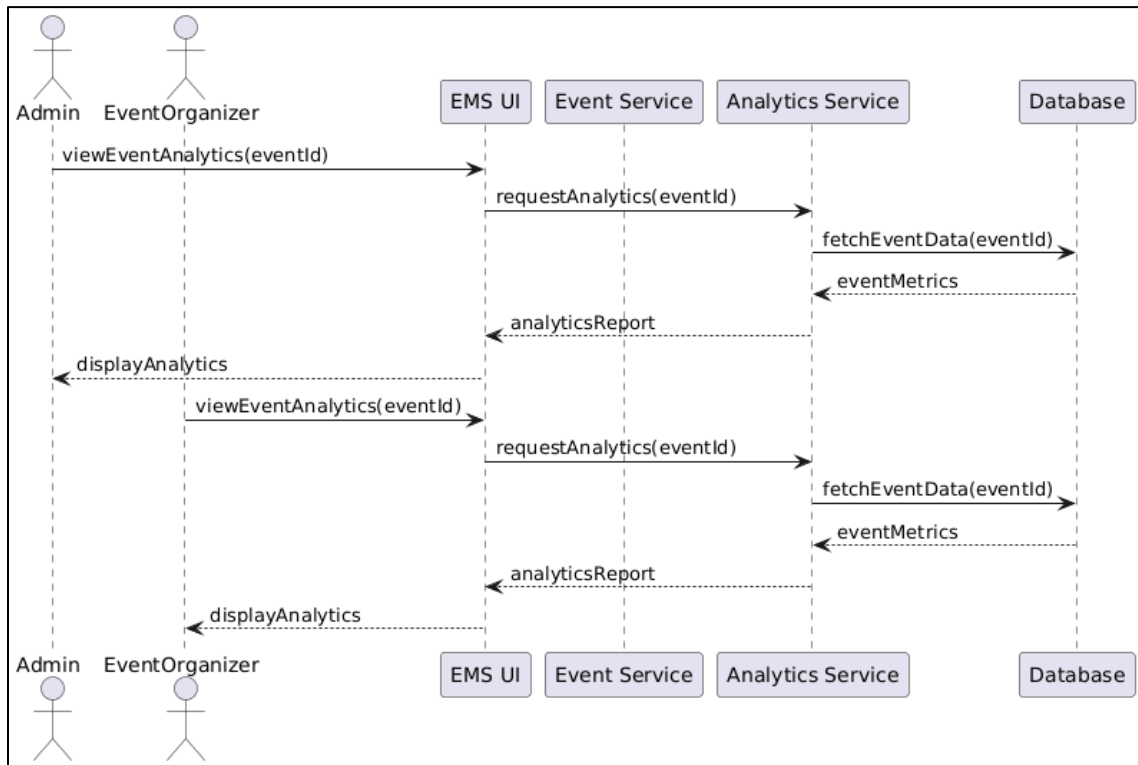


Figure 4.20: Event Analytics Generation

## 4.7 Interface Design

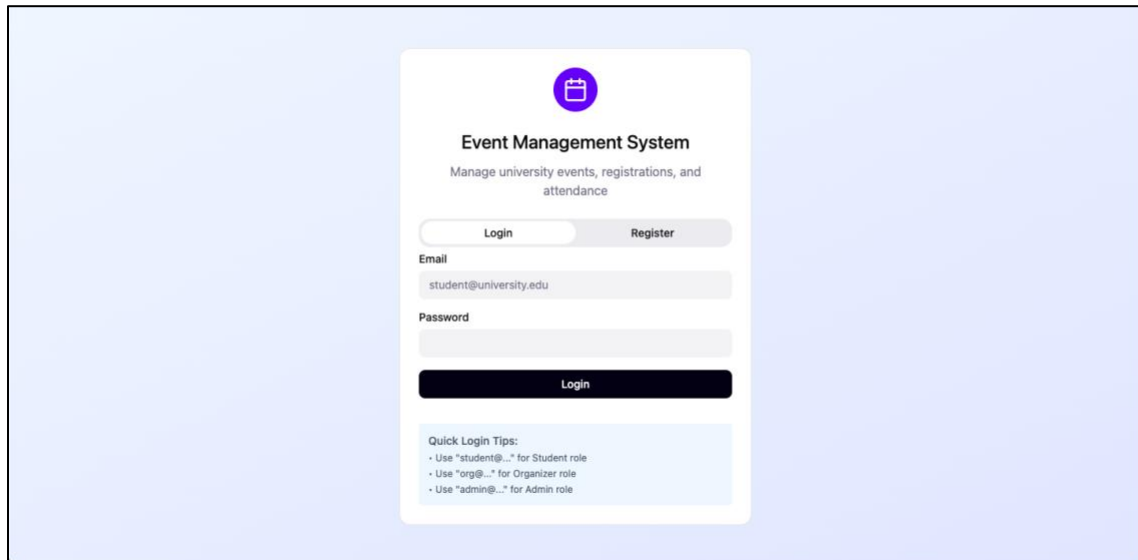
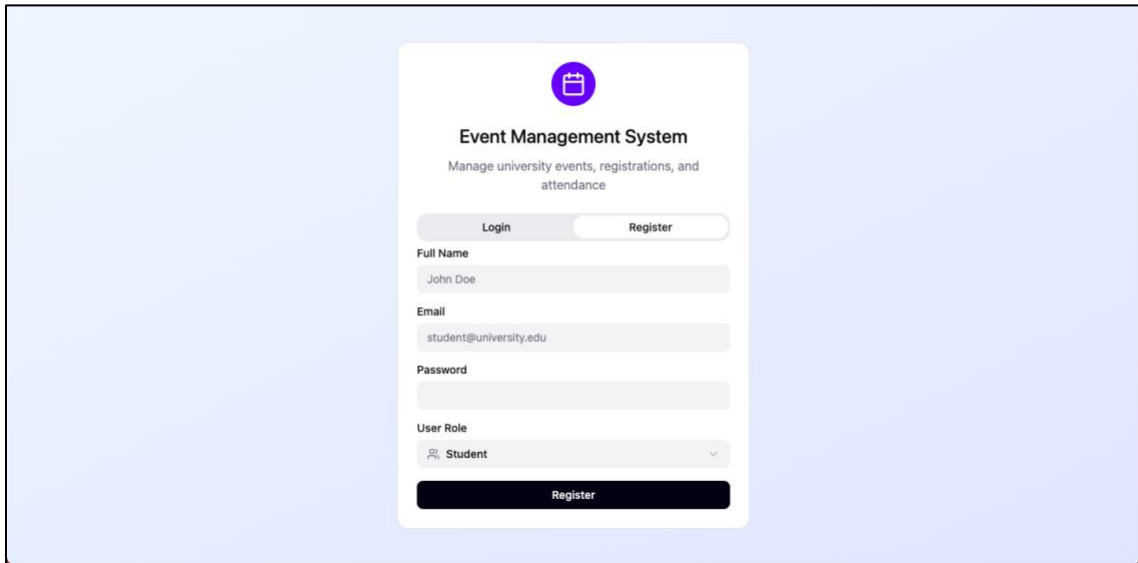
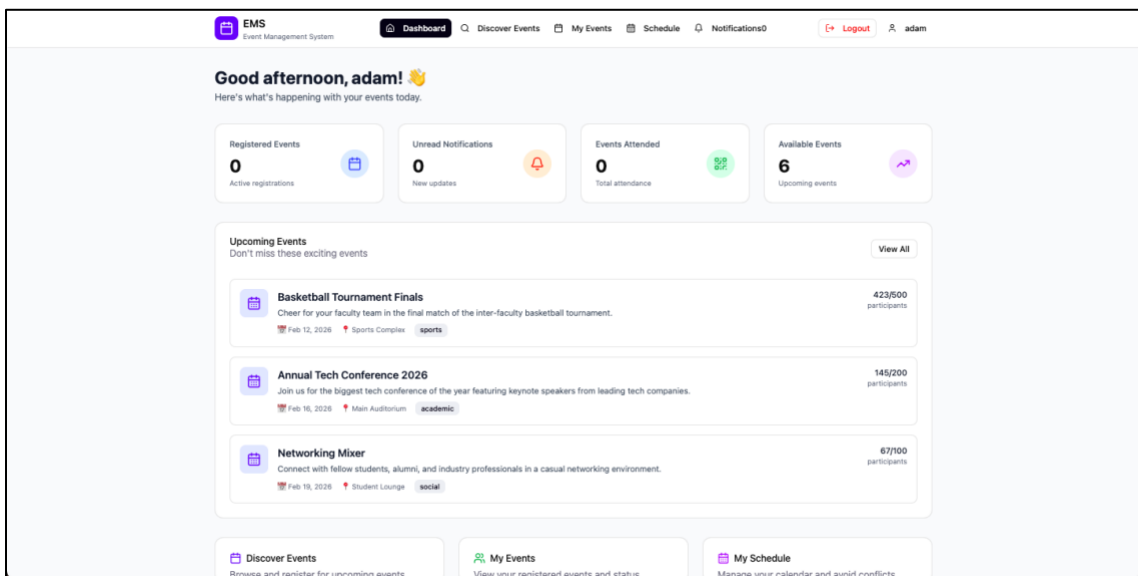


Figure 4.21: EMS Login Page



The image shows the registration page of the Event Management System (EMS). It features a central white card on a light blue background. At the top of the card is a purple calendar icon. Below the icon, the text reads "Event Management System" and "Manage university events, registrations, and attendance". There are two tabs: "Login" and "Register", with "Register" being the active tab. The registration form includes fields for "Full Name" (with the placeholder "John Doe"), "Email" (with the placeholder "student@university.edu"), "Password", and "User Role" (a dropdown menu currently showing "Student"). A large black "Register" button is at the bottom of the card.

Figure 4.22: EMS Register Page



The image shows the student dashboard of the Event Management System (EMS). The dashboard has a header with the EMS logo, navigation links (Dashboard, Discover Events, My Events, Schedule, Notifications), and a user profile section (Logout, adam). The main content area starts with a greeting "Good afternoon, adam!" and a sub-header "Here's what's happening with your events today." Below this are four summary cards: "Registered Events" (0 Active registrations), "Unread Notifications" (0 New updates), "Events Attended" (0 Total attendance), and "Available Events" (6 Upcoming events). The "Upcoming Events" section follows, listing three events: "Basketball Tournament Finals" (Feb 12, 2026, Sports Complex, 423/500 participants), "Annual Tech Conference 2026" (Feb 16, 2026, Main Auditorium, 145/200 participants), and "Networking Mixer" (Feb 18, 2026, Student Lounge, 67/100 participants). At the bottom, there are three action cards: "Discover Events" (Browse and register for upcoming events), "My Events" (View your registered events and status), and "My Schedule" (Manage your calendar and avoid conflicts).

Figure 4.23: EMS Student Dashboard

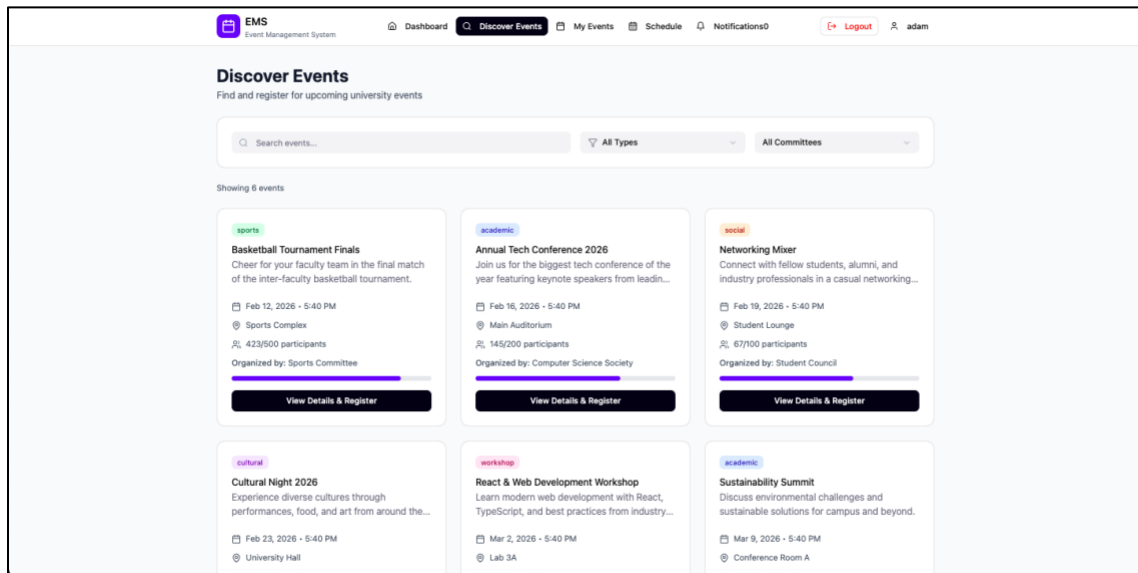


Figure 4.24: EMS Student Event Search

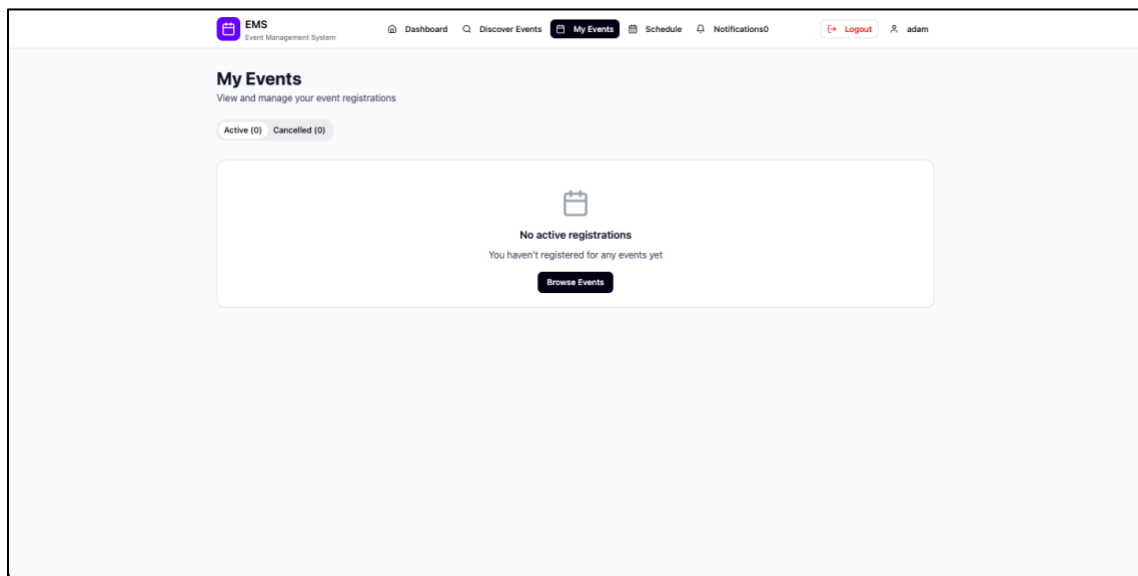


Figure 4.25: EMS Student Personal Events List

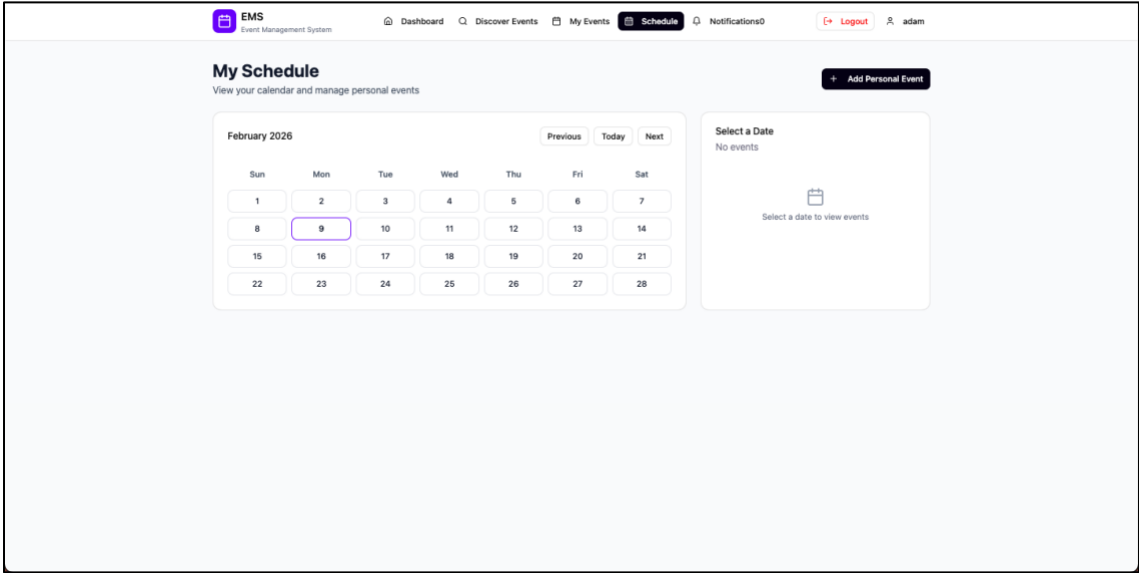


Figure 4.26: EMS Student Personal Schedule

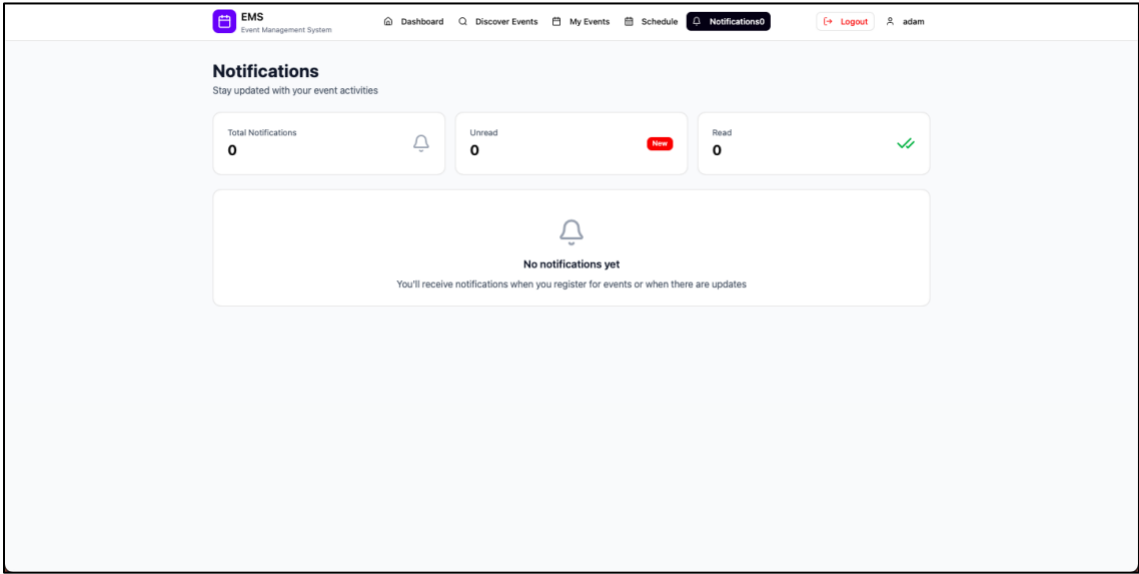


Figure 4.27: EMS Student Notifications List

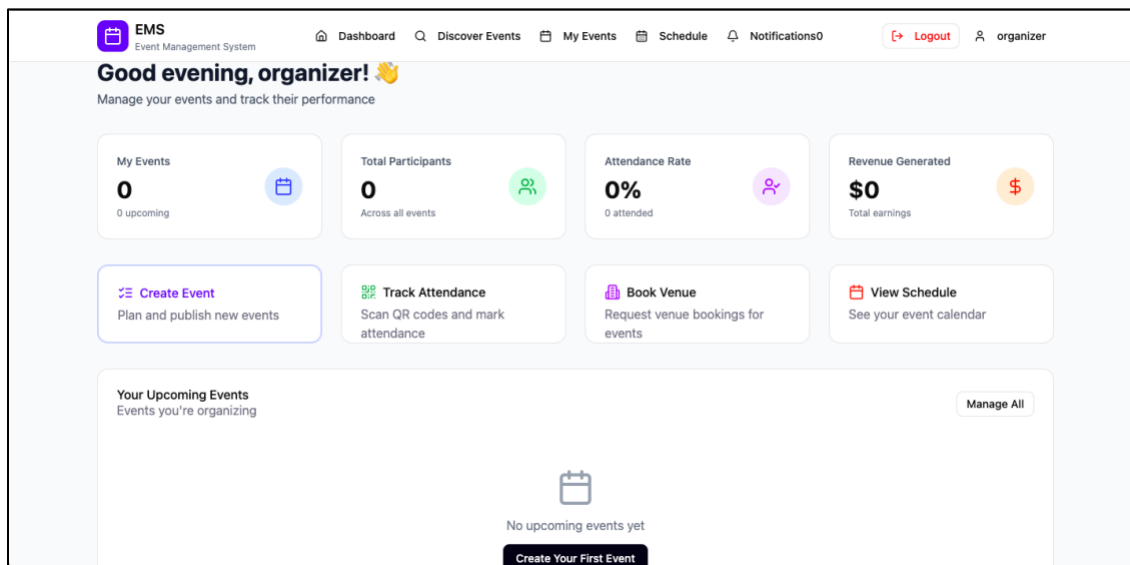


Figure 4.28: EMS Organizer Dashboard

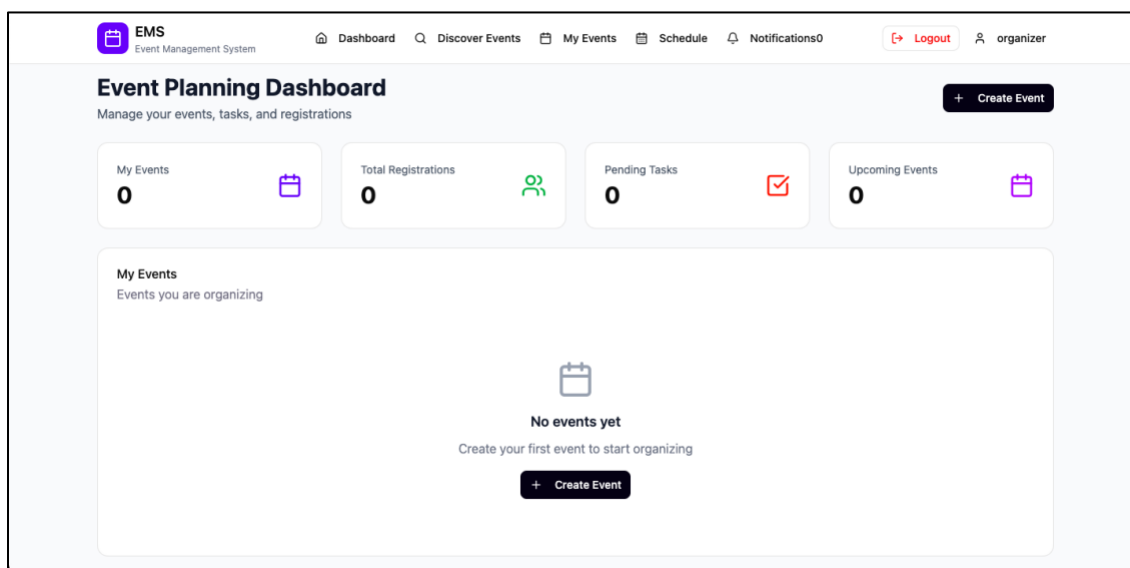


Figure 4.29: EMS Organizer Event Planning Dashboard



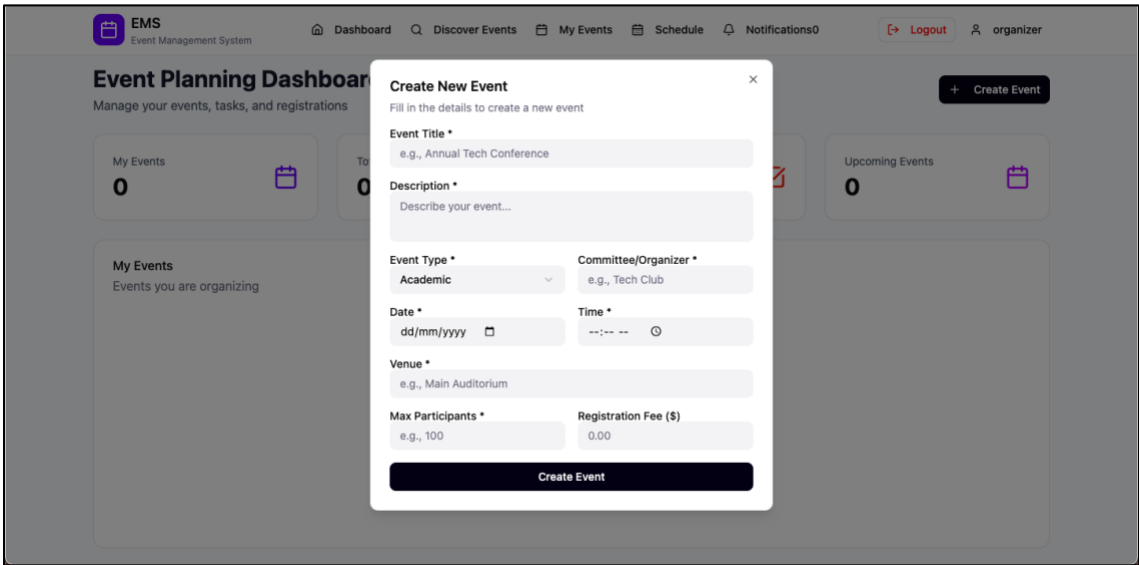


Figure 4.30: EMS Organizer Event Creation Window

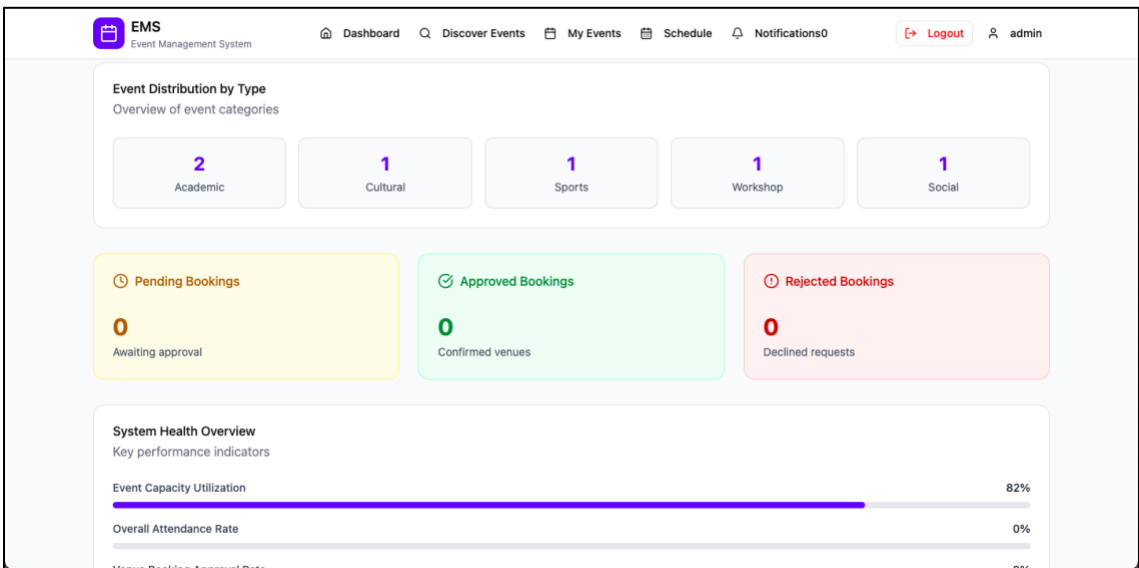


Figure 4.31: EMS Admin Dashboard

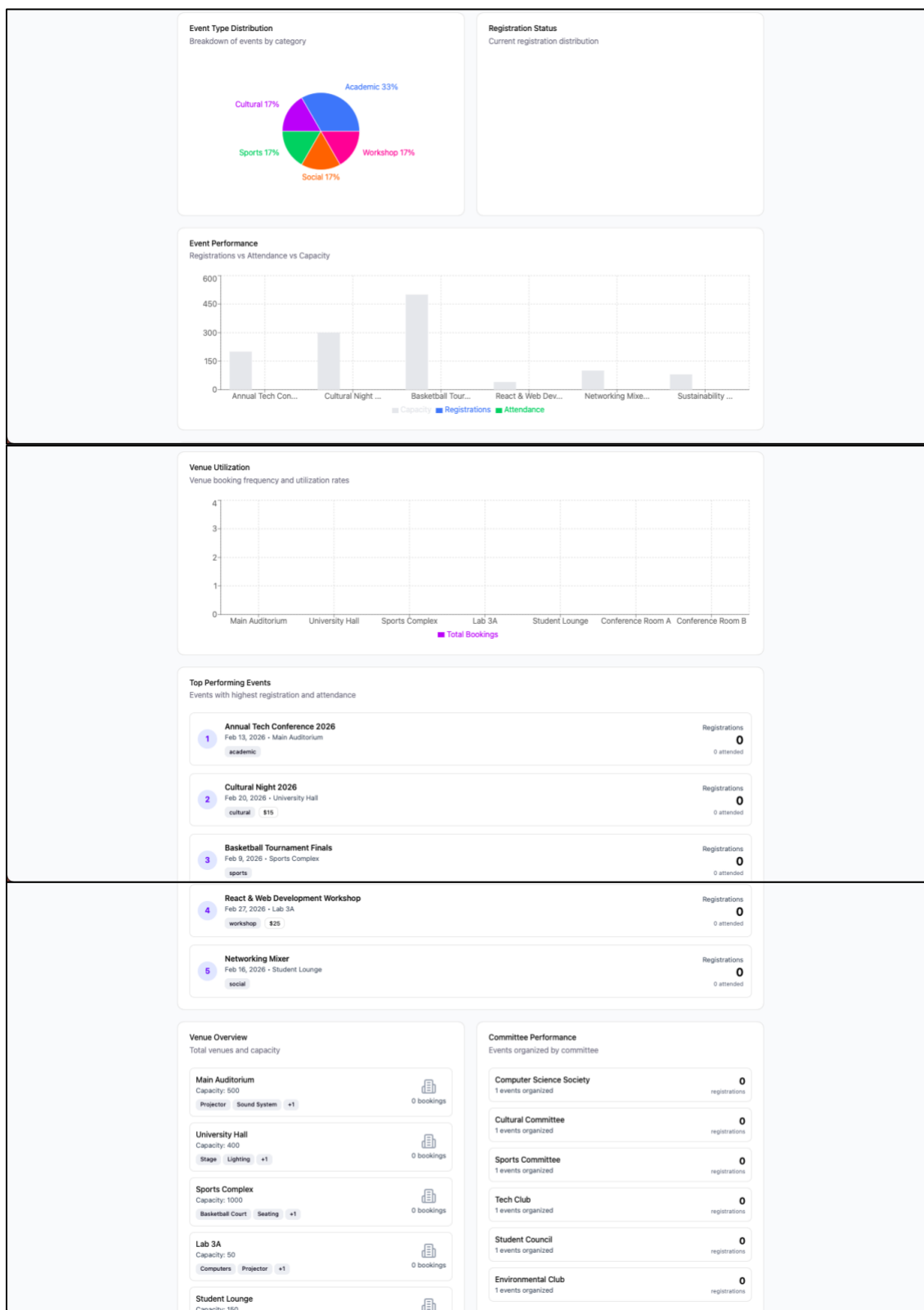


Figure 4.32: EMS Admin Analytics Dashboard

## 4.8 Summary

This chapter presented the system design of the EMS, derived from the findings of the survey and the interviews. The issues identified in existing university event management practices, including fragmented event discovery, unclear registration processes, scheduling clashes, and inefficient venue management, were translated into structured system designs.

The rich picture, use case, and activity diagrams were used to model current problems and proposed workflows. The class and sequence diagrams defined the system structure and key interactions, while the interface diagram illustrated role-based user access and navigation. Overall, the system design ensures that the EMS addresses the identified requirements and provides a clear foundation for system implementation in the next chapter.

## Chapter 5: Implementation

### 5.1 Activities in FYP2

The activities to be done in FYP2 will focus on implementing the EMS based on the system design provided in the previous chapter. This will include the development of backend and frontend, database setup and system integration.

### 5.2 FYP2 Milestones

#### 5.2.1 Milestones

Table 5.1: Milestones for FYP2

| Week  | Milestones                                                   | Due Date   |
|-------|--------------------------------------------------------------|------------|
| 1     | Review of FYP1 Project Work                                  | 07/04/2026 |
| 2-3   | Update Plan and Requirements                                 | 21/04/2026 |
| 3-5   | Database design and backend setup                            | 28/04/2026 |
| 5-8   | Implementation of core backend modules                       | 19/05/2026 |
| 9     | Progress review                                              | 25/05/2026 |
| 10-12 | Frontend development and integration of backend and frontend | 16/06/2026 |
| 13    | System testing and bug fixing                                | 23/06/2026 |

|       |                                               |               |
|-------|-----------------------------------------------|---------------|
| 13-14 | User acceptance testing and final refinements | 30/06/2026    |
| 14    | Final Draft of Report                         | 30/06/2026    |
| 15    | Final Report Submission                       | 07/07/2026    |
| 16-17 | Presentation                                  | Mid July 2026 |

## Chapter 6: Conclusion

Event management processes in MMU are scattered, unstandardized, and require laborious effort just to accomplish basic tasks. The proposed EMS is intended to reverse the dynamic so that no matter what their university role is, members of the university will find it significantly easier to perform their required tasks. Based on survey and interview findings, it has been determined that the EMS will need to involve users of three types based on their role: students, event organizers, and administrators. Respectively, the tasks of the users in general will be to register for events, to create events, and to manage events. Further elaboration regarding the structure of the EMS is given in Chapter 4 and will assist in the development and implementation of the system during FYP2.

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## Appendix A



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### **CPT6314 Final Year Project (FYP) 1 Meeting Log Trimester OCT / NOV 2025 (Trimester ID:2530)**

|                                                                                                                   |                                                           |
|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| <b>Meeting Date:</b><br>26 November 2025                                                                          | <b>Meeting No.:</b><br>1                                  |
| <b>Meeting Mode:</b><br>Face-to-Face                                                                              |                                                           |
| <b>Project ID:</b><br>FYP01-SE-T2530-0794                                                                         | <b>Project Type:</b><br>Application-based                 |
| <b>Project Title :</b><br>Event Management System with Automated Scheduling Optimization Using Genetic Algorithms |                                                           |
| <b>Student ID :</b><br>1221101160                                                                                 | <b>Student Name:</b><br>Muhammad Nabil Naufal bin Md Zaid |
| <b>Student Programme and Specialisation:</b><br>Bachelor of Computer Science (Hons.) (Software Engineering)       |                                                           |
| <b>Supervisor Name:</b><br>Mohana A/P Muniandy, Dr.                                                               | <b>Co-Supervisor Name:</b><br>(If applicable)             |
| <b>Collaborating Company:</b><br>(If applicable)                                                                  | <b>Company Supervisor Name:</b><br>(If applicable)        |

### Appendix A 1: Meeting Log 1 Page 1

**1. WORK DONE**

**Tasks:** Problem Formulation and Project Planning / Background Study or Literature Review / Requirement Analysis or Theoretical Framework / Design or Research Methodology / Prototype Development or Proof of Concept / Draft Report Completion

**Details (in point form):**

I have partially completed Chapter 1: Introduction of the Interim Report from Subsection 1.1 Overview to Subsection 1.5 Project Scope.

**Appendix A 2: Meeting Log 1 Page 2**

**2. WORK TO BE DONE**

**Tasks:** Problem Formulation and Project Planning / Background Study or Literature Review / Requirement Analysis or Theoretical Framework / Design or Research Methodology / Prototype Development or Proof of Concept/ Draft Report Completion

**Details (in point form):**

I have been instructed to complete Chapter 1: Introduction and then proceed with Chapter 2: Literature Review of the Interim Report. For this, I will have to secure and go through resources on past event management systems, scheduling, genetic algorithms, and the Kanban system

**Appendix A 3: Meeting Log 1 Page 3**



### 3. PROBLEMS ENCOUNTERED AND SOLUTIONS

I have been instructed to abandon the Software Development Life Cycle (SDLC) in favour of the Kanban system as the latter would be more appropriate for a one-man project such as this one.

Subsection 1.3 Objectives was written unclear due to my use of abbreviations in it. For that reason, I have been instructed to edit it to make it clearer by including the full forms of the abbreviations in the subsection.

Within Subsection 1.5 Project Scope, I included details on the Main Users and their Functions for my Event Management System which was deemed inappropriate for this part of the report. I have removed it from this subsection for that reason.

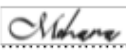
Chapter 1: Introduction has not been completed as per the Final Year Project Handbook. I have been instructed to complete it.

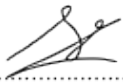
## Appendix A 4: Meeting Log 1 Page 4

**3. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)**

Continue working on chapter 2.

Appendix A 5: Meeting Log 1 Page 5

  
.....  
Supervisor's Signature

  
.....  
Student's Signature

.....  
Co-Supervisor's Signature  
(if applicable)

.....  
Company Supervisor's Signature  
(if applicable)

**IMPORTANT NOTES TO STUDENTS:**

1. Items 1 – 3 are to be completed by the students prior to the meeting. Item 4 is to be completed by the supervisor / co-supervisor / company supervisor.
2. Student has to upload the soft copies of the meeting logs in eBwise and also attach them along with interim (FYP1) report.  
Minimum requirement is SIX Meeting Logs (Period: Week 3 to Week 12). Students can have fortnightly meetings with the supervisor.
3. Log sheets provide the basis for evaluating the General Effort (Project Management, Attitude, and Technical Competency) of the student, by the supervisor and also for checking the attendance requirement of the student, by the FYP Committee.  
  
This also provide the student with feedback from the supervisor / co-supervisor / company supervisor on the tasks done and provide the plan for the upcoming tasks. This can provide the motivation for the student to give consistent and efficient effort throughout the period of FYP.
4. Student who fails to meet the minimum requirement (six nos.) of log sheets will not be allowed to submit FYP report.

Appendix A 6: Meeting Log 1 Page 6



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**CPT6314 Final Year Project (FYP) 1 Meeting Log  
Trimester OCT / NOV 2025 (Trimester ID:2530)**

|                                                                                                                   |                                                           |
|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| <b>Meeting Date:</b><br>3 December 2025                                                                           | <b>Meeting No.:</b><br>2                                  |
| <b>Meeting Mode:</b><br>Face-to-Face                                                                              |                                                           |
| <b>Project ID:</b><br>FYP01-SE-T2530-0794                                                                         | <b>Project Type:</b><br>Application-based                 |
| <b>Project Title :</b><br>Event Management System with Automated Scheduling Optimization Using Genetic Algorithms |                                                           |
| <b>Student ID :</b><br>1221101160                                                                                 | <b>Student Name:</b><br>Muhammad Nabil Naufal bin Md Zaid |
| <b>Student Programme and Specialisation:</b><br>Bachelor of Computer Science (Hons.) (Software Engineering)       |                                                           |
| <b>Supervisor Name:</b><br>Mohana A/P Muniandy, Dr.                                                               | <b>Co-Supervisor Name:</b><br>(if applicable)             |
| <b>Collaborating Company:</b><br>(if applicable)                                                                  | <b>Company Supervisor Name:</b><br>(if applicable)        |

Appendix A 7: Meeting Log 2 Page 1

**1. WORK DONE**

**Tasks:** Problem Formulation and Project Planning / Background Study or Literature Review / Requirement Analysis or Theoretical Framework / Design or Research Methodology / Prototype Development or Proof of Concept / Draft Report Completion

**Details (in point form):**

I have completed Chapter 1 and Chapter 2 as per the supervisor's advice and guide.

**Appendix A 8: Meeting Log 2 Page 2**

**2. WORK TO BE DONE**

~~Tasks: Problem Formulation and Project Planning / Background Study or Literature Review / Requirement Analysis or Theoretical Framework / Design or Research Methodology / Prototype Development or Proof of Concept/ Draft Report Completion~~

*(Please strike out the tasks, which are not applicable)*

**Details (in point form):**

I must complete a part of Chapter 3: Requirements Analysis as advised by the supervisor, specifically with subsections 3.1 Overview, 3.2 Fact-Finding Techniques and 3.3 Requirement.

**Appendix A 9: Meeting Log 2 Page 3**

**3. PROBLEMS ENCOUNTERED AND SOLUTIONS**

*[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]*

I had written the Methodology section wrong where I explained what the Kanban framework is instead of what I intend to do with it. Hence, I must correct the Methodology section so that I instead elaborate on what I plan to do with it whereby I explain in more detail what my plan is in each of the Kanban stages I have set.

The sources I cited are too old. I must add citations to sources that confirm the content of my report but are also as recent as 2019.

For Subsection 2.5 Existing Systems for Event Management, I did not refer to any actual existing system. I must now do so.

**3. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)**

For any technical resources, the citation cannot be more than 5 years ago. Therefore, use sources that are 2021 onwards.

For chapter 3, design the questionnaire and interview questions with their objectives clearly stated.



  
.....  
Supervisor's Signature

  
.....  
Student's Signature

.....  
Co-Supervisor's Signature  
(if applicable)

.....  
Company Supervisor's Signature  
(if applicable)

**IMPORTANT NOTES TO STUDENTS:**

1. Items 1 – 3 are to be completed by the students prior to the meeting. Item 4 is to be completed by the supervisor / co-supervisor / company supervisor.
2. Student has to upload the soft copies of the meeting logs in eBwise and also attach them along with interim (FYP1) report.  
Minimum requirement is SIX Meeting Logs (Period: Week 3 to Week 12). Students can have fortnightly meetings with the supervisor.
3. Log sheets provide the basis for evaluating the General Effort (Project Management, Attitude, and Technical Competency) of the student, by the supervisor and also for checking the attendance requirement of the student, by the FYP Committee.  
  
This also provide the student with feedback from the supervisor / co-supervisor / company supervisor on the tasks done and provide the plan for the upcoming tasks. This can provide the motivation for the student to give consistent and efficient effort throughout the period of FYP.
4. Student who fails to meet the minimum requirement (six nos.) of log sheets will not be allowed to submit FYP report.



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**CPT6314 Final Year Project (FYP) 1 Meeting Log**  
**Trimester OCT / NOV 2025 (Trimester ID:2530)**

|                                                                                                                   |                                                           |
|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| <b>Meeting Date:</b><br>10 December 2025                                                                          | <b>Meeting No.:</b><br>3                                  |
| <b>Meeting Mode:</b><br>Face-to-Face                                                                              |                                                           |
| <b>Project ID:</b><br>FYP01-SE-T2530-0794                                                                         | <b>Project Type:</b><br>Application-based                 |
| <b>Project Title :</b><br>Event Management System with Automated Scheduling Optimization Using Genetic Algorithms |                                                           |
| <b>Student ID :</b><br>1221101160                                                                                 | <b>Student Name:</b><br>Muhammad Nabil Naufal bin Md Zaid |
| <b>Student Programme and Specialisation:</b><br>Bachelor of Computer Science (Hons.) (Software Engineering)       |                                                           |
| <b>Supervisor Name:</b><br>Dr. Mohana A/P Muniandy                                                                | <b>Co-Supervisor Name:</b><br>(if applicable)             |
| <b>Collaborating Company:</b><br>(if applicable)                                                                  | <b>Company Supervisor Name:</b><br>(if applicable)        |

**1. WORK DONE**

*[Please write the details of the work done, after the last meeting]*

**Tasks:** Problem Formulation and Project Planning / Background Study or Literature Review / Requirement Analysis or Theoretical Framework / Design or Research Methodology / Prototype Development or Proof of Concept / Draft Report Completion

***(Please strike out the tasks, which are not applicable)***

**Details (in point form):**

Questionnaire for Students for survey in subsection 3.2.2 Questionnaire Design.

## Appendix A 14: Meeting Log 3 Page 2

**2. WORK TO BE DONE**

*[Please write the details of the work to be done, before the next meeting]*

**Tasks:** ~~Problem Formulation and Project Planning / Background Study or Literature Review / Requirement Analysis or Theoretical Framework / Design or Research Methodology / Prototype Development or Proof of Concept / Draft Report Completion~~

***(Please strike out the tasks, which are not applicable)***

**Details (in point form):**

The questionnaire for respondents of other university roles such as Club Members and Event Organizers.

**3. PROBLEMS ENCOUNTERED AND SOLUTIONS**

*[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]*

In various parts of the report, the author used 'I' to refer to himself. All instances of which will subsequently be changed to 'the author'.

Subsection 2.5 Existing Systems for Event Management does not contain any reference to a real existing Event Management System due to them being locked a paywall. The author is advised to look for people's reviews of existing Event Management Systems.

Subsection 1.6 Methodology discusses matters beyond the scope of the subsection and chapter. The author is given interview methods as an example of something suitable to be included in the subsection.

**Appendix A 16: Meeting Log 3 Page 4**

**4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)**

Start collecting data after the survey questions have been designed and approved by me.

.....  
Supervisor's Signature.....  
Student's Signature.....  
Co-Supervisor's Signature  
(if applicable).....  
Company Supervisor's Signature  
(if applicable)



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**CPT6314 Final Year Project (FYP) 1 Meeting Log**  
**Trimester OCT / NOV 2025 (Trimester ID:2530)**

|                                                                                                                   |                                                           |
|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| <b>Meeting Date:</b><br>18 December 2025                                                                          | <b>Meeting No.:</b><br>4                                  |
| <b>Meeting Mode:</b><br>Face-to-Face                                                                              |                                                           |
| <b>Project ID:</b><br>FYP01-SE-T2530-0794                                                                         | <b>Project Type:</b><br>Application-based                 |
| <b>Project Title :</b><br>Event Management System with Automated Scheduling Optimization Using Genetic Algorithms |                                                           |
| <b>Student ID :</b><br>1221101160                                                                                 | <b>Student Name:</b><br>Muhammad Nabil Naufal bin Md Zaid |
| <b>Student Programme and Specialisation:</b><br>Bachelor of Computer Science (Hons.) (Software Engineering)       |                                                           |
| <b>Supervisor Name:</b><br>Dr. Mohana A/P Muniandy                                                                | <b>Co-Supervisor Name:</b><br>(if applicable)             |
| <b>Collaborating Company:</b><br>(if applicable)                                                                  | <b>Company Supervisor Name:</b><br>(if applicable)        |

**1. WORK DONE**

*[Please write the details of the work done, after the last meeting]*

**Tasks:** ~~Problem Formulation and Project Planning / Background Study or Literature Review / Requirement Analysis or Theoretical Framework / Design or Research Methodology / Prototype Development or Proof of Concept / Draft Report Completion~~

***(Please strike out the tasks, which are not applicable)***

**Details (in point form):**

Completed Questionnaire and Interview questions for users of all university roles.

## Appendix A 19: Meeting Log 4 Page 2



**2. WORK TO BE DONE**

*[Please write the details of the work to be done, before the next meeting]*

**Tasks:** ~~Problem Formulation and Project Planning / Background Study or Literature Review / Requirement Analysis or Theoretical Framework / Design or Research Methodology / Prototype Development or Proof of Concept / Draft Report Completion~~

*(Please strike out the tasks, which are not applicable)*

**Details (in point form):**

I have to perform requirements elicitation by conducting surveys and interviews with the appropriate individuals.

**3. PROBLEMS ENCOUNTERED AND SOLUTIONS**

*[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]*

- Change question type with scalable answers to 'Likert Scale' or 'Semantic Scale' as appropriate.
- For questions with multiple choices, separate each answer with a comma, not a forward slash.
- For questions with multiple choices and multiple answers, add "(Select all that apply)" at the end of all questions that apply.
- Add "Other" answer option where people who take the survey can add their own answer to a question.
- Add "Why?" to certain questions as appropriate.

**4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)**

Make the necessary changes to improve your questionnaire. Choose your interviewee to include someone who has experience organizing events.

.....  
.. Supervisor's Signature.....  
Student's Signature.....  
Co-Supervisor's Signature  
(if applicable).....  
Company Supervisor's Signature  
(if applicable)



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**CPT6314 Final Year Project (FYP) 1 Meeting Log  
Trimester OCT / NOV 2025 (Trimester ID:2530)**

|                                                                                                                   |                                                           |
|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| <b>Meeting Date:</b><br>7 January 2025                                                                            | <b>Meeting No.:</b><br>5                                  |
| <b>Meeting Mode:</b><br>Face-to-Face                                                                              |                                                           |
| <b>Project ID:</b><br>FYP01-SE-T2530-0794                                                                         | <b>Project Type:</b><br>Application-based                 |
| <b>Project Title :</b><br>Event Management System with Automated Scheduling Optimization Using Genetic Algorithms |                                                           |
| <b>Student ID :</b><br>1221101160                                                                                 | <b>Student Name:</b><br>Muhammad Nabil Naufal bin Md Zaid |
| <b>Student Programme and Specialisation:</b><br>Bachelor of Computer Science (Hons.) (Software Engineering)       |                                                           |
| <b>Supervisor Name:</b><br>Dr. Mohana A/P Muniandy                                                                | <b>Co-Supervisor Name:</b><br>(if applicable)             |
| <b>Collaborating Company:</b><br>(if applicable)                                                                  | <b>Company Supervisor Name:</b><br>(if applicable)        |

**1. WORK DONE**

*[Please write the details of the work done, after the last meeting]*

**Tasks:** ~~Problem Formulation and Project Planning / Background Study or Literature Review / Requirement Analysis or Theoretical Framework / Design or Research Methodology / Prototype Development or Proof of Concept / Draft Report Completion~~

***(Please strike out the tasks, which are not applicable)***

**Details (in point form):**

- I have conducted a survey for which I have received 30 responses.
- I have conducted interviews with five individuals, one of which was Event Organizer and the other four Students.
- I have partially completed Subsection 3.2.2.1: Analysis on Results.

## Appendix A 24: Meeting Log 5 Page 2

**2. WORK TO BE DONE**

*[Please write the details of the work to be done, before the next meeting]*

**Tasks:** ~~Problem Formulation and Project Planning / Background Study or Literature Review / Requirement Analysis or Theoretical Framework / Design or Research Methodology / Prototype Development or Proof of Concept/ Draft Report Completion~~

*(Please strike out the tasks, which are not applicable)*

**Details (in point form):**

- I plan to make one more interview with a Committee Member.
- I have to complete the analyses of the survey responses and the interviews.
- Upon completion of the analyses, I will proceed with Chapter 4: System Design.

**3. PROBLEMS ENCOUNTERED AND SOLUTIONS**

*[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]*

For my partially complete analysis of the survey responses, I made it that I first summarized the findings and then added the related figures beneath the summary. I must modify it so that each figure is given their own relevant paragraph instead.

**4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)**

After the analyses of your interview and survey, write about the functional requirements, non-functional requirements and user requirements before proceeding to chapter 4.

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Supervisor's Signature.....  
Student's Signature.....  
Co-Supervisor's Signature  
(if applicable).....  
Company Supervisor's Signature  
(if applicable)





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**CPT6314 Final Year Project (FYP) 1 Meeting Log**  
**Trimester OCT / NOV 2025 (Trimester ID:2530)**

|                                                                                                                   |                                                           |
|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| <b>Meeting Date:</b><br>30 January 2026                                                                           | <b>Meeting No.:</b><br>6                                  |
| <b>Meeting Mode:</b><br>Online                                                                                    |                                                           |
| <b>Project ID:</b><br>FYP01-SE-T2530-0794                                                                         | <b>Project Type:</b><br>Application-based                 |
| <b>Project Title :</b><br>Event Management System with Automated Scheduling Optimization Using Genetic Algorithms |                                                           |
| <b>Student ID :</b><br>1221101160                                                                                 | <b>Student Name:</b><br>Muhammad Nabil Naufal bin Md Zaid |
| <b>Student Programme and Specialisation:</b><br>Bachelor of Computer Science (Hons.) (Software Engineering)       |                                                           |
| <b>Supervisor Name:</b><br>Dr. Mohana A/P Muniandy                                                                | <b>Co-Supervisor Name:</b><br>(if applicable)             |
| <b>Collaborating Company:</b><br>(if applicable)                                                                  | <b>Company Supervisor Name:</b><br>(if applicable)        |

**1. WORK DONE**

*[Please write the details of the work done, after the last meeting]*

**Tasks:** ~~Problem Formulation and Project Planning / Background Study or Literature Review / Requirement Analysis or Theoretical Framework / Design or Research Methodology / Prototype Development or Proof of Concept / Draft Report Completion~~

***(Please strike out the tasks, which are not applicable)***

**Details (in point form):**

- Completed Subsection 3.3 Requirements.
- Completed Chapter 4: System Design.

## Appendix A 29: Meeting Log 6 Page 2

**2. WORK TO BE DONE**

*[Please write the details of the work to be done, before the next meeting]*

**Tasks:** ~~Problem Formulation and Project Planning / Background Study or Literature Review / Requirement Analysis or Theoretical Framework / Design or Research Methodology / Prototype Development or Proof of Concept/ Draft Report Completion~~

***(Please strike out the tasks, which are not applicable)***

**Details (in point form):**

- Complete Subsection 3.4 Summary in Chapter 3: Requirements Analysis.
- Complete Chapter 5: Implementation Plan
- Complete Chapter 6: Conclusion
- Complete Draft Report.

### Appendix A 30: Meeting Log 6 Page 3

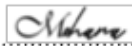
**3. PROBLEMS ENCOUNTERED AND SOLUTIONS**

*[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]*

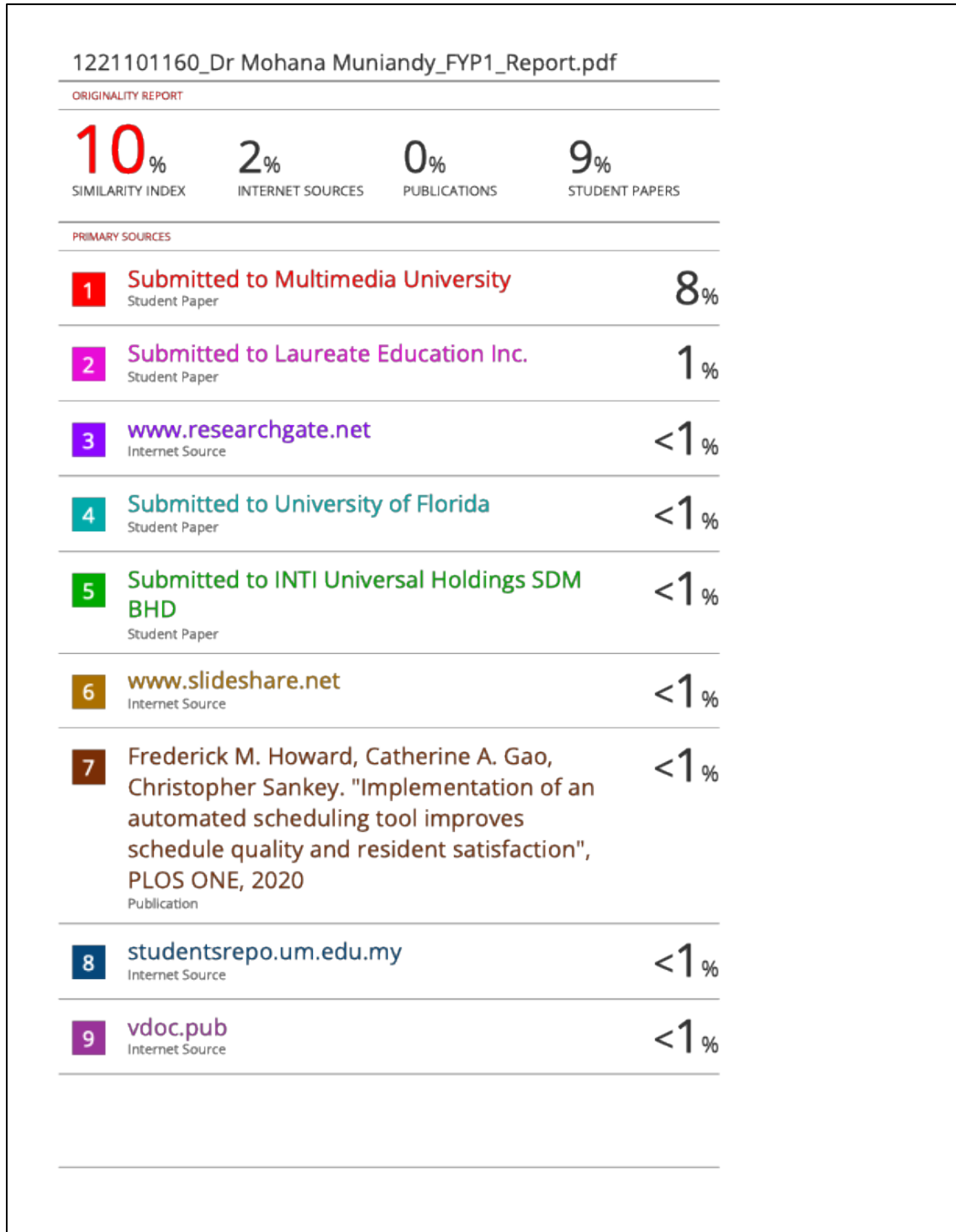
- Many of the user requirements are in fact functional requirements. Replace them all with user requirements.
- Multiple rich picture diagrams for the system are provided. Combine all the diagrams into one.
- In the use case diagram, many lines overlap. Rearrange the actors and lines so that no overlapping occurs.
- The diagrams are not centralized. Centralize all the diagrams.
- In the class diagram, functions are not provided, and unnecessary classes are added. Add functions to classes and remove unnecessary classes such as PaymentProof.
- In Figure 4.17, payment proof logic is included. Remove payment proof logic.
- In Subsection 4.7, interface design is done incorrectly. Provide hypothetical UI with use of Figma.

**4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)**

Work on completing chapter 4. Start writing chapter 5 and 6.

.....  
Supervisor's Signature.....  
Student's Signature.....  
Co-Supervisor's Signature  
(if applicable).....  
Company Supervisor's Signature  
(if applicable)

## Appendix B



Appendix B 1: Turnitin Similarity Index

## Appendix C

### FYP1 Student Interview Transcript

#### Question 1: Can you walk through the last campus event you attended, from discovering it to attending it?

Muadz: It was a badminton competition.

Nazim: I discovered this event due to a lecturer recommending it. When I reached this place, I had to register. The registration was cumbersome, taking 10 minutes. We needed lots of information about ourselves for registration. I even had to give identification cards.

Nelly: If I'm not mistaken, it was a STEM event by Sekretariat Rakan Muda (SRM). My friend who was in the committee told me about it.

Hesham: It was a Career Fair that the university organized. It wasn't on campus. They sent an email about it and we had to apply through a website. After that they sent a confirmation email and we had to scan a QR code for the transportation to get there. And they sent the reply by email the night before.

#### Question 2: How did you register and how long did it take?

Muadz: I registered through a link that my friend sent me. That link brought me to a form with which I registered. However I was notified only a few days later that I had been registered.

Nazim: I registered using Microsoft Forms. It took around 10-15 minutes and it required my Student ID.

Nelly: I didn't register for anything because I didn't join any activities and just watched it. But it probably wouldn't have taken that long.

Hesham: They sent us a website and we had to register with our email through that website and then get the confirmation, and we had to register for transportation. It was really convoluted. We weren't sure whether we were registered or not.

### Q3: What tools or platforms were involved?

Muadz: Instagram, social media, Outlook email, and Whatsapp through which my friend sent me the link.

Nazim: Microsoft Forms.

Nelly: I think they mainly used Google Forms but I did not register for anything so I'm not sure.

Hesham: Career fair's website, Google/Microsoft Forms for the transportation, and they sent them to me through email.

### Question 4: Was there any part of the process that you found annoying?

Muadz: It was the fact that my registration confirmation email could get lost in emails because my email is used for schoolwork too.

Nazim: I would it's probably the registration process. The event finding can be annoying too as often, I would find an event and it's already concluded. It's also really hard to know whether an event has started or ended.

Nelly: I'd hate it when I'm registering and I need to pay by QR on the phone. I would like it for me to direct me straight to the payment gateway.

Hesham: The transportation one, because there's limited slots, so you'd expect them to send a confirmation message as to whether we're in or not. We did get a confirmation message it came very late. Also, we had to use an external website to register which made things confusing.

### Question 5: What problems do you usually face when joining campus events?

Muadz: Being informed. This is because they often inform us on Instagram accounts I don't follow or notify us on notice boards that I don't check out.

Nazim: Sometimes I forget my ID card, making it harder to join events. There should be another way for me to register.

## Appendix C 2: FYP Student Interview Transcript Page 2



Nelly: It's not a problem but I always expect more from events. Like they often oversell their events.

Hesham: I'd say visibility is the main problem because there's no standard way to access or see the events. The main way we have now is WhatsApp groups mainly. I know someone who made a website for listing all events but it was made by scraping Instagram for the events.

### **Question 6: Have you ever missed an event you wanted to attend? Why?**

Muadz: Yeah, it was a fun run that I wanted to join. I didn't know it was happening so I didn't get to register for it.

Nazim: Yes, often I see the event poster and the event has already concluded. I think it's because the time is only on the poster, nowhere else on MMU portal etc.

Nelly: Always, because I always fail to notice the announcement about the event, because there's no centralized place for campus event notices. Even if there is, not all campus events are covered.

Hesham: Oh plenty. And every time it was because I didn't know it was happening.

### **Question 7: When things go wrong (no confirmation, confusion), how do you usually handle it?**

Muadz: Usually I just be patient and wait until I get notified.

Nazim: Most of the time, I just anxiously wait.

Nelly: I would try to contact the person in charge, because it's better to do that than to make assumptions.

Hesham: I actually had to email back 12 hours before the event asking if there would be confirmation and then they replied. The main communication platform would be email but not much else.

## **Appendix C 3: FYP Student Interview Transcript Page 3**

### Question 8: Have you experienced event clashes with classes or other events?

Muadz: Yes. The event I talked about lasted for a whole day and took the time of three classes. So I had to miss all those classes.

Nazim: Many times. Sometimes I schedule meetings/assignments at the same time as events because I just screenshot events without adding them to my calendar.

Nelly: Absolutely and I feel dreaded every time.

Hesham: With other events, no, because I don't attend that many. But with classes, not too many times because I don't attend that many events. A lot of the times, the events fall on weekends so it's fine. But the ones on campus sometimes fall during classes. Some of them tell you you'll get an exemption letter, some don't even mention it, so you're not sure if you can attend it without ruining your attendance.

### Question 9: How do you currently check if an event clashes with your timetable?

Muadz: I wait until they notify me because they usually don't tell when the event will be until the day of the event.

Nazim: If I remember correctly, I compare it with my own calendar. If I forget about it, I just don't.

Nelly: I cross-check manually.

Hesham: I look at the event date and time and I look at my class schedule.

### Question 10: How do organizers usually communicate changes or updates?

Muadz: I haven't entered any event that had changes so I don't know.

Nazim: Usually via email.

## Appendix C 4: FYP Student Interview Transcript Page 4

Nelly: They would update on their Instagram story or announce it on WhatsApp. If the participant is still registered, they would most likely send an email about the update.

Hesham: By email or WhatsApp groups they add us to.

### Question 11: Do you feel informed enough before and during events?

Muadz: I wish they would establish a date beforehand. Sometimes they do and sometimes they don't. I also wish they would have a WhatsApp group for event updates.

Nazim: Sometimes I miss those emails so I come to the campus and the event's at another time so I check my emails and I find out the event's time's been changed. It's more a problem with me than with email. But emails also sometimes don't give notifications.

Nelly: Sometimes yes and sometimes no. It depends on how they update their events. Sometimes their updates would get lost among my messages because they give their updates in random WhatsApp groups.

Hesham: It depends on the organizers, but a lot of the time the information is a bit hidden as well so you have to go looking for it, or ask people like the organizers and they'd provide the info.

### Question 12: If you could design the “perfect event app” for students, what must it have? Why?

Muadz: A place where students can poll or choose the date of the event that satisfies the majority. One place for event reminders and updates. And a calendar feature.

Nazim: I would like it to automatically add events I like to my Google calendar, and add notifications of any events happening. Also something to make registration easier (all information being there in one place).

Nelly: Maybe a calendar and a dashboard for events we're interested to join, similar to the wishlist on the Shoppee app.

Hesham: It must have a calendar timeline thing that shows all events happening on campus. It must also have filtering tools filtering by club events, music events and so on. If it's an event management platform, you can have groups on it where it adds you to groups when you've registered for the event. And also there should a forum where people can ask about the event

## Appendix C 5: FYP Student Interview Transcript Page 5

beforehand. It should also support registering to the events and all that stuff. There should also be a history of events attended for users.

### Question 13: What would make you actually use such a system instead of WhatsApp/Instagram?

Muadz: If I were to join many events, if the app has many good features, and if my friends had it.

Nazim: It would have to provide easy event registration and also enable communication with the organizer and so on.

Nelly: Maybe if the system has attendance tracking and convenience of payment method.

Hesham: I would rather use a system than any of the other fragmented methods mentioned. That would lessen the problem of visibility as all the events are contained in one place.

### Question 14: Would you trust the system for official certificates or attendance tracking?

Muadz: If it was made by MMU, yes.

Nazim: If it's certified by the university, yes. If many clubs are using it, yes.

Nelly: Yes.

Hesham: It depends on how it's done. If it stores the information I'm giving to the organizers then that's iffy. For example, if you're applying for events, and they ask for address, the university doesn't need that. Essential information like name and email are fine.

### Question 15: What would make you stop using an Event Management System even if it exists?

Muadz: If I were too busy with my studies and I had no event to join, or too many of the events I wanted to join clashed with my classes.

Nazim: If it started giving me wrong times for events, or random events that I don't join keep popping up.

Nelly: Probably if the interface is confusing.

Hesham: There should be a standard way for creating events. If it's too restrictive for only small events and prioritizes only big events, that would make me stop using it.

## Appendix C 7: FYP Student Interview Transcript Page 7

## FYP1 Event Organizer Interview Transcript

### Question 1: Can you explain how event approvals are handled in your department?

Dr Ng Kok Why: Normally, the procedure for organizing the activities or events in MMU is that the club or society come up with the proposal and in the last page of the proposal, the secretary need to sign followed by their society treasurer and then the president. After that, they will go to a second layer which is the adviser who will see through the proposal and sign it. This is followed by the STYLE office staff followed by the STYLE office director. The adviser and the STYLE staff have to make sure the event don't break rules (e.g. doesn't touch on sensitive issue and doesn't involve hurting someone).

### Q2: What rules must clubs comply with before approval?

Dr Ng Kok Why: Usually, the club will list down the purpose and objectives of the event and the members involved, the audience involved and the budget required to support the event.

### Q3: How many event requests do you manage in a typical semester?

Dr Ng Kok Why: It's very subjective. Early in the year, there are usually not that many. Many events will be held later in the year or close to the end of the year. If I had to say, every month would have about 3-4 events.

### Q4: What becomes difficult when volume increases?

Dr Ng Kok Why: When the number of events increases, the first problem would be there not being enough venues. If the club and event aren't big, they could use a classroom. But if it's a big event, there might problems where they couldn't book big venues like MPH and DTC.

### Q5: How do you track:

- Event status?

- Venue usage?
- Attendance?

Dr Ng Kok Why: Usually if a club or society wants to book a venue, they will have to refer to the STyLE office. It's for faculty staff where our admin staff will check through the system the venues one by one.

### Q6: What information do you wish you had instantly but don't?

Dr Ng Kok Why: Usually, that would be the venue and number of students. When organizing events, the hardest thing to predict would be the number of attendees. This makes it hard to give the suitable venue. For example, if an event were allocated a huge venue but the number of attendees is small, that wouldn't be good. The same would be true if an event were allocated a small venue but more people came than expected.

### Q7: What risks worry you most in student-organized events?

Dr Ng Kok Why: If there's a running competition for example, and a student has not enough stamina, but they still force themselves to compete, that would be a risk.

### Q8: How do you ensure compliance with university policies?

Dr Ng Kok Why: Usually, if an event involves high risk, they would have to contact the ambulance or police to stand by. For example, if there was a fun run, they would have ask an ambulance to stand by in case an emergency happens. But some clubs overlook this policy.

### Q9: What frustrates you most about the current event management process?

Dr Ng Kok Why: The system isn't smart enough to allocate the suitable venue to events. For example, allocating a big venue for events with not that many attendees.

## Appendix C 9: FYP Event Organizer Interview Transcript Page 2

**Q10: Where do challenges usually happen?**

Dr Ng Kok Why: It would usually be in determining how many people would turn up for events. Sometimes people would say they would turn up for an event but then not show up.

**Q11: What would convince you to adopt a new Event Management System?**

Dr Ng Kok Why: If the event management system could help optimize facilities and the time for the venue, I would welcome this event management system.

**Q12: What would block its implementation?**

Dr Ng Kok Why: If you were to develop an event management system and it's very efficient, it would still have to go through a few departments who will go through your system and double-check it and make sure it's secure. And then you'd need to go to top management for approval.

**Q13: Ideally, how should event management work in 3–5 years?**

Dr Ng Kok Why: The event management system would be able to help us quickly book the suitable venue where the system would help us tick all the necessary equipments which will help us contact the person in charge for arranging all the items.

**Q14: How could data or automation help your department?**

Dr Ng Kok Why: If your system can learn from previous data from previous experiences, for example, someone has booked an event for 50 people, and the system would give the suitable venue.